

विद्यया ऽ मृतमश्नुते



एन सी ई आर टी
NCERT



NEET STUDY MATERIAL (NEW)



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PERIODIC TABLE

① History:

- **Dobereiner**: Law of triads → $\boxed{\text{Li, Na, K}}$ $\boxed{\text{Ca, Sr, Ba}}$ $\boxed{\text{Cl, Br, I}}$
- **Newland**: Law of octaves (properties repeat after 8th element)
 valid upto calcium
- **Lothar Meyer**: Lothar Meyer curve → Atomic volume vs Atomic weight
- **Mendeleev**: Mendeleev periodic table → Based on increasing atomic weight
- **Henry Moseley**: Modern periodic table → Based on increasing atomic number

Atomic mass:

$$\frac{1^{st} + 3^{rd}}{2} = 2^{nd}$$



- Left to right
• along period → $Z_{eff} \uparrow$, size $\downarrow$, IE $\uparrow$, EA $\uparrow$, EN $\uparrow$
Non metallic/acidic chr $\uparrow$, Metallic/basic chr $\downarrow$
- Top to bottom
• along group → $Z_{eff} \downarrow$, size $\uparrow$, no of shell(n) $\uparrow$, IE $\downarrow$, EA $\downarrow$, EN $\downarrow$
Non metallic/acidic chr $\downarrow$, Metallic/basic chr $\uparrow$

② Electronic configuration

- Think about Atomic number of previous inert gas of given element/given atomic number
- Subtraction: $\text{Given atomic number} - \text{previous inert gas atomic no.} = \text{Remaining number}$

$S \rightarrow f \rightarrow d \rightarrow p$ me put kuo.

#					* Inert gas
n=1	S				² He
n=2	S	P			¹⁰ Ne
n=3	S	P			¹⁸ Ar
n=4	S	d	P		³⁶ Kr
n=5	S	d	P		⁵⁴ Xe
n=6	S	f	d	P	⁸⁶ Rn
n=7	S	f	d	P	¹¹⁸ Og

f-block

- * 58 → 71 → Lanthanide series
- * 90 → 103 → Actinide series

$$\{ ns^2 \rightarrow (n-2)f^{14} \rightarrow (n-1)d^{10} \rightarrow np^6 \}$$

→ practice:

• AN: 65

• AN: 48

S-block → No exception

Group 1 (Alkali Metals)	Group 2 (Alkaline Earth Metals)
Symbol of element	Symbol of element
Electronic Configuration	Electronic Configuration
Lithium	Beryllium
Sodium	Magnesium
Potassium	Calcium
Rubidium	Strontium
Cesium	Barium
Francium	Radium

HaLiNa Ki Rb
Ce Fryad

Beta Mange Car
Scooter Bap Razi

D-block 3d-series

Scandium	Sc	21	[Ar] 3d ¹ 4s ²
Titanium	Ti	22	[Ar] 3d ² 4s ²
Vanadium	V	23	[Ar] 3d ³ 4s ²
Chromium	Cr	24	[Ar] 3d ⁵ 4s ¹
Manganese	Mn	25	[Ar] 3d ⁵ 4s ²
Iron	Fe	26	[Ar] 3d ⁶ 4s ²
Cobalt	Co	27	[Ar] 3d ⁷ 4s ²
Nickel	Ni	28	[Ar] 3d ⁸ 4s ²
Copper	Cu	29	[Ar] 3d ¹⁰ 4s ¹
Zinc	Zn	30	[Ar] 3d ¹⁰ 4s ²

Sun Tu Vivah Crle Mujhse Fer Coi Nahi Cahega Zanu

5d-series

Hafnium	Hf	72	[Xe] 4f ¹⁴ 5d ² 6s ²
Tantalum	Ta	73	[Xe] 4f ¹⁴ 5d ³ 6s ²
Tungsten	W	74	[Xe] 4f ¹⁴ 5d ⁴ 6s ²
Rhenium	Re	75	[Xe] 4f ¹⁴ 5d ⁵ 6s ²
Osmium	Os	76	[Xe] 4f ¹⁴ 5d ⁶ 6s ²
Iridium	Ir	77	[Xe] 4f ¹⁴ 5d ⁷ 6s ²
Platinum	Pt	78	[Xe] 4f ¹⁴ 5d ⁹ 6s ¹
Gold	Au	79	[Xe] 4f ¹⁴ 5d ¹⁰ 6s ¹
Mercury	Hg	80	[Xe] 4f ¹⁴ 5d ¹⁰ 6s ²

La Hafta Tau Warna Re Os Iron rod se
Pitai Aur Hogi

4d-series

Rhodium	Rh	45	[Kr] 4d ⁸ 5s ¹
Palladium	Pd	46	[Kr] 4d ¹⁰
Silver	Ag	47	[Kr] 4d ¹⁰ 5s ¹
Cadmium	Cd	48	[Kr] 4d ¹⁰ 5s ²
Indium	In	49	[Kr] 4d ¹⁰ 5s ² 5p ¹
Tin	Sn	50	[Kr] 4d ¹⁰ 5s ² 5p ²
Antimony	Sb	51	[Kr] 4d ¹⁰ 5s ² 5p ³
Tellurium	Te	52	[Kr] 4d ¹⁰ 5s ² 5p ⁴
Iodine	I	53	[Kr] 4d ¹⁰ 5s ² 5p ⁵
Xenon	Xe	54	[Kr] 4d ¹⁰ 5s ² 5p ⁶

Ye Zindigi Nahi Mohobbat Tumhari Ru Rokr Pukadegi
Ag Chandni

P-block → No exception

Element	Symbol	Atomic Number	Electronic Configuration
Boron	B	5	[He] 2s ² 2p ¹
Carbon	C	6	[He] 2s ² 2p ²
Nitrogen	N	7	[He] 2s ² 2p ³
Oxygen	O	8	[He] 2s ² 2p ⁴
Fluorine	F	9	[He] 2s ² 2p ⁵
Neon	Ne	10	[He] 2s ² 2p ⁶

Baigan Aalu Gajar In Thaila

Element	Symbol	Atomic Number	Electronic Configuration
Lithium	Li	3	[He] 2s ¹
Sodium	Na	11	[Ne] 3s ¹
Potassium	K	19	[Ar] 4s ¹
Rubidium	Rb	37	[Kr] 5s ¹
Cesium	Cs	55	[Xe] 6s ¹
Francium	Fr	87	[Rn] 7s ¹

Cahe Siv Ge Sune Parbati

Element	Symbol	Atomic Number	Electronic Configuration
Beryllium	Be	4	[He] 2s ²
Magnesium	Mg	12	[Ne] 3s ²
Calcium	Ca	20	[Ar] 4s ²
Strontium	Sr	38	[Kr] 5s ²
Barium	Ba	56	[Xe] 6s ²
Radium	Ra	88	[Rn] 7s ²

New Zealand Pakistan Australia Sb Buzy

Element	Symbol	Atomic Number	Electronic Configuration
Boron	B	5	[He] 2s ² 2p ¹
Carbon	C	6	[He] 2s ² 2p ²
Nitrogen	N	7	[He] 2s ² 2p ³
Oxygen	O	8	[He] 2s ² 2p ⁴
Fluorine	F	9	[He] 2s ² 2p ⁵
Neon	Ne	10	[He] 2s ² 2p ⁶

Os Se Tepo

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Strontium	Sr	38	[Kr] 5s ²
Barium	Ba	56	[Xe] 6s ²
Radium	Ra	88	[Rn] 7s ²

Hena Neena Aur Karina ka X-ray Rangeen

f-block

Lanthanide series

58 → 71

Element	Symbol	Atomic Number	Electron Configuration
Lanthanum	La	57	$[Xe] 4f^0 5d^1 6s^2$
Cerium	Ce	58	$[Xe] 4f^1 5d^1 6s^2$
Praseodymium	Pr	59	$[Xe] 4f^3 6s^2$
Neodymium	Nd	60	$[Xe] 4f^4 6s^2$
Europium	Eu	61	$[Xe] 4f^7 6s^2$
Gadolinium	Gd	62	$[Xe] 4f^7 5d^1 6s^2$
Terbium	Tb	63	$[Xe] 4f^9 6s^2$
Dysprosium	Dy	64	$[Xe] 4f^{10} 6s^2$
Hoium	Ho	65	$[Xe] 4f^{11} 6s^2$
Erbium	Er	66	$[Xe] 4f^{12} 6s^2$
Thulium	Tm	67	$[Xe] 4f^{13} 6s^2$
Ytterbium	Yb	68	$[Xe] 4f^{14} 6s^2$
Lutetium	Lu	69	$[Xe] 4f^{14} 5d^1 6s^2$

Actinide series

90 → 103

Element	Symbol	Atomic Number	Electron Configuration
Actinium	Ac	89	$[Rn] 5f^0 6d^1 7s^2$
Thorium	Th	90	$[Rn] 5f^0 6d^2 7s^2$
Protactinium	Pa	91	$[Rn] 5f^2 6d^1 7s^2$
Uranium	U	92	$[Rn] 5f^3 6d^1 7s^2$
Neptunium	Np	93	$[Rn] 5f^4 6d^1 7s^2$
Plutonium	Pu	94	$[Rn] 5f^6 6d^1 7s^2$
Americium	Am	95	$[Rn] 5f^7 6d^1 7s^2$
Curium	Cm	96	$[Rn] 5f^7 6d^2 7s^2$
Berkelium	Bk	97	$[Rn] 5f^9 7s^2$
Californium	Cf	98	$[Rn] 5f^{10} 7s^2$
Einsteinium	Es	99	$[Rn] 5f^{11} 7s^2$
Fermium	Fm	100	$[Rn] 5f^{12} 7s^2$
Mendelevium	Md	101	$[Rn] 5f^{13} 7s^2$
Nobelium	No	102	$[Rn] 5f^{14} 7s^2$
Lawrencium	Lr	103	$[Rn] 5f^{14} 6d^1 7s^2$

Cene Par Nadiya Prem se
Samai, Eu Gadgad Tb Dil
Hoa Ere Tm Ybi Lutlo

Taro Par Unki Nazar Pari
Asman Chamka Bizli Carki
Esan Faramos Manchali Nikli
Lrki

③

IUPAC

101 → 118

Digit	Name	Abbreviation
0	nil	n
1	un	u
2	bi	b
3	tri	t
4	quad	q
5	pent	p
6	hex	h
7	sept	s
8	oct	o
9	non	n

• 120



Unbinilium

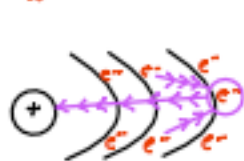
ium → prefix

Atomic Number	Name according to IUPAC nomenclature	Symbol	IUPAC Official Name	IUPAC Symbol
101	Ununium	Uun	Mendelevium	Md
102	Unbium	Uub	Nobelium	No
103	Untrium	Uut	Lawrencium	Lr
104	Unquadium	Uuq	Rutherfordium	Rf
105	Unpentium	Uup	Dubnium	Db
106	Unhexium	Uuh	Seaborgium	Sg
107	Unseptium	Uus	Solmium	Bh
108	Unoctium	Uuo	Hassium	Hs
109	Unnennium	Uun	Mitnerium	Mt
110	Undecium	Uud	Darmstadtium	Ds
111	Unduanium	Udu	Röntgenium	Rg
112	Unium	Uub	Copernicium	Cn
113	Untrium	Uut	Nihonium	Nh
114	Unquadium	Uuq	Flerovium	Fl
115	Unpentium	Uup	Moscovium	Mc
116	Unhexium	Uuh	Livermorium	Lv
117	Unseptium	Uus	Tennessine	Ts
118	Unoctium	Uuo	Oganesson	Og

Mendelev get Nobel in Lawrence, Rutherford dube sea me Bohr has
mat Dar se Roye Copper , Neha Fly to Mosco to Live Ten Og

* Periodic trends

④ Effective nuclear charge (Z_{eff})



atomic number
 $Z_{eff} = Z - \sigma$
 Net attraction / Net contraction
 Screening constant ($s > p > d > f$)

$$Z_{eff} \propto \frac{1}{r} \propto \frac{1}{size}$$

poor shielding

Actinide contraction (poor $5f$) > Lanthanide contraction (poor $4f$) > Transitional contraction (poor $3d$)

⑤ Size

(a) Atomic radius

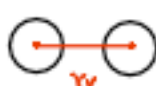
(i) Covalent radius



(ii) Metallic radius



(iii) Vander Waals radius



$$\therefore r_v > r_m > r_c \rightarrow r_v = 2r_c$$

(b) Ionic radius

$$R_i \propto \frac{-ve \text{ (anionic charge)}}{+ve \text{ (cationic charge)}}$$

* Different ion of same element: $M^- > M > M^+$

* Isoelectronic species: $N^{3-} > O^{2-} > F^- > Na^+ > Mg^{2+}$

(c) Blockwise size

(i) s-block

Group wise: No exception ($T \rightarrow B$: size $\uparrow$)



(ii) p-block

13 th	14 th	15 th	16 th	17 th	18 th
B	C	N	O	F	Ne
Al	Si	P	S	Cl	Ar
Ga	Ge	As	Se	Br	Kr
In	Sn	Sb	Te	I	Xe
Tl	Pb	Bi	Po	At	

No exception
 Transitional contraction

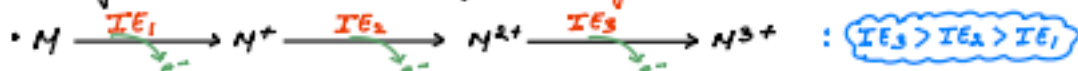
(iii) d-block

Groupwise size:



⑥ Ionisation Energy: Energy required to remove electron from isolated gaseous atom.

Always Endothermic ($\Delta H = +ve$) $\rightarrow$ Always

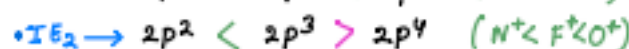


$IE \propto Z_{eff} \propto \frac{1}{size} < \text{penetration} < \text{stability}$
 $ns^2 > np^1$ (Half/fully filled)
 $np^3 > np^4$ (Half/filled)
 $np^6 > np^5$

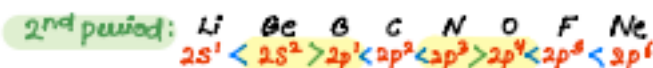
* Final order: $ns^1 < ns^2 > np^1 < np^2 < np^3 > np^4 < np^5 < np^6$

* For 2nd/3rd IE: Remove reqd electrons & re-write configuration.

$\Rightarrow$ for eg: Find 2nd IE for N, O, F?



(a) period-wise IE



(b) Blockwise IE

(i) s-block

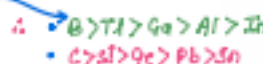
No exception
 $IE \propto \frac{1}{size}$

(ii) p-block

13 th	14 th	15 th	16 th	17 th	18 th
B	C	N	O	F	Ne
Al	Si	P	S	Cl	Ar
Ga	Ge	As	Se	Br	Kr
In	Sn	Sb	Te	I	Xe
Tl	Pb	Bi	Po	At	

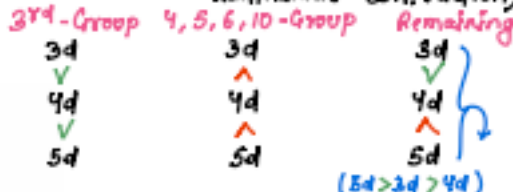
Transitional contraction

Lanthanide contraction



(iii) d-block

(irregular variation due to Lanthanide contraction)



⑦ Electron gain Enthalpy: addition of e^- : $M + e^- \longrightarrow M^-$

- Endothermic $\rightarrow \Delta H_{eg} = +ve$ * Note the $-ve \Delta H_{eg}$, higher the tendency to gain e^-
- Exothermic $\rightarrow \Delta H_{eg} = -ve$ $\therefore$ Higher electron affinity.

$\rightarrow$ Neutral species $\xrightarrow{e^-} -\Delta H$: exothermic (except: N, Be, Mg, Inert gas) $\rightarrow +\Delta H$

$\rightarrow$ Cation (M^+) $\xrightarrow{e^-} -\Delta H$: exothermic (all)

$\rightarrow$ anion (M^-) $\xrightarrow{e^-} +\Delta H$: Endothermic (all)

{ * electron should be added one by one }

$\rightarrow EA \propto Z_{eff} \propto \frac{1}{\text{size}} \propto \text{penetration} \propto \frac{1}{\text{stability}}$

$ns^1 > np^1$

Half filled: $np^2 > np^3$

fully filled: $EA = 0$ (inert gas)

* Maximum EA in periodic table
 $\downarrow$
Chlorine

Final order: (EA) $ns^1 > ns^2 < np^1 < np^2 > np^3 < np^4 < np^5 > np^6$

Penetration

Half Filled

fully filled

(a) periodwise EA

2nd period: Li Be B C N O F Ne
 $2s^1 > 2s^2 < 2p^1 < 2p^2 > 2p^3 < 2p^4 < 2p^5 > 2p^6$

$\rightarrow Ne < Be < N < B < Li < C < O < F$ (NB NB LIC OFFICE)

3rd period: Na Mg Al Si P S Cl Ar
 $3s^1 > 3s^2 < 3p^1 < 3p^2 > 3p^3 < 3p^4 < 3p^5 > 3p^6$

$\rightarrow Ar < Mg < Al < Na < P < Si < S < Cl$ (Ar MAN Padhega SSC)

(b) Group wise

2nd period: B C N O F
3rd period: Al Si P S Cl

$\rightarrow$ EA of 3rd period greater than 2nd period
 $\rightarrow$ b/c of e^-e^- repulsion in sp subshell

* Imp $\rightarrow$ 16th group: S > Se > Te > Po > O
order $\rightarrow$ 17th group: Cl > F > Br > I

⑧ Electronegativity: Tendency of atom to attract shared pair of e^-

$A^{5-} \leftarrow \leftarrow \leftarrow \leftarrow B^{5+}$
(More EN) $\rightarrow$ polar bond $\rightarrow$ due to EN difference (ΔEN)

Group (Period II)	Li	Be	B	C	N	O	F
Electronegativity	1.0	1.5	2.0	2.5	3.0	3.5	4.0
Group (Period III)	Na	Mg	Al	Si	P	S	Cl
Electronegativity	1.0	1.2	1.5	1.8	2.1	2.5	3.0

$\rightarrow EN \propto Z_{eff} \propto \frac{1}{\text{size}} \propto \text{oxidation state} \propto \%s \text{ character} \propto \text{Bond Polarity} \propto \text{Acidic character} \propto \frac{1}{\text{Basic chr}}$

Same element can show different OS

$\therefore$ different EN

$Mn^{+2} < Mn^{+4} < Mn^{+7}$
 $MnO < MnO_2 < Mn_2O_7$

$sp > sp^2 > sp^3$
 $50\%s \quad 33\%s \quad 25\%s$

ionic chr.
 $\downarrow$
from EN value

⑨ Oxides:

(a) Metallic oxide

Generally basic

eg: $K_2O, MgO, Pb_2O_3 \dots$

Metallic oxide $\xrightarrow{H_2O}$ Base

$\{ Na_2O \xrightarrow{H_2O} NaOH \}$

(b) Non metallic oxide

Generally Acidic

eg: $NO_2, Cl_2O_3 \dots$

* Non metallic oxide $\xrightarrow{H_2O}$ Acid

$\{ CO_2 \xrightarrow{H_2O} H_2CO_3 \}$

* few Non-metallic oxides are Neutral: CO, NO, H_2O, N_2O

(c) Amphoteric oxide

act as acid & base

Zn Be Al Sb As
Pb Sn V^{+5} Ga Cr^{+3}

Cr^{+3} = Ampho
 Cr^{+6} = Acidic

(d) d-block oxide

variable oxidation state

* Change Nature
 $\rightarrow +1, +2, +3$: Basic
 $\rightarrow +4$: Ampho
 $\rightarrow +5, +6, +7$: Acidic

⑩ Application:

a) Ionisation energy:

IE_1	IE_2	IE_3	IE_4
165 eV	196 eV	656 eV	596 eV

no of valance e^-

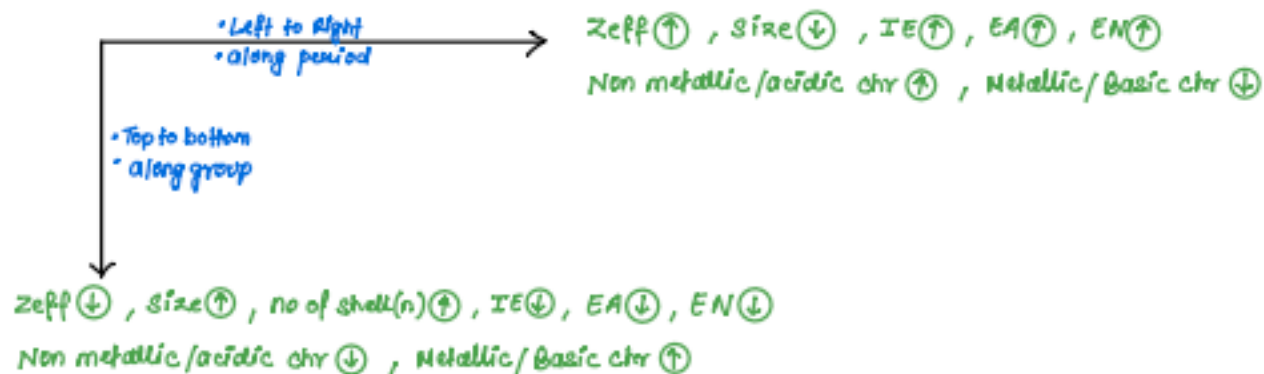
• Valance e^- : 2

• OS: +2

b) electronegativity

$\Delta EN \propto$ non-metallic character $\propto$ % ionic character $\propto$ Acidic chr / Basic chr

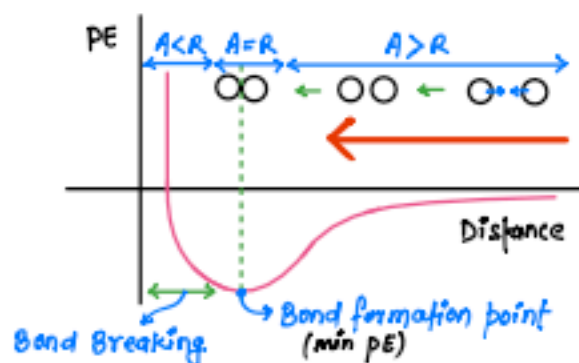
⑪ General trends:



CHEMICAL BONDING

① Chemical bond :

- Force of attraction b/w atom, molecules, ions etc
- Bond formation $\rightarrow$ always exothermic ($-\Delta H$)
- Bond dissociation $\rightarrow$ always endothermic ($+\Delta H$)
- why bond form $\rightarrow$ to acquire Maximum stability & minimum potential energy.



Types of chemical Bond

Strong Bonds (more stable)

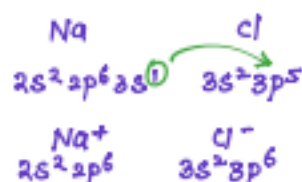
- Ionic bond
- Covalent bond
- Co-ordinate bond
- Metallic bond

Weak Bonds

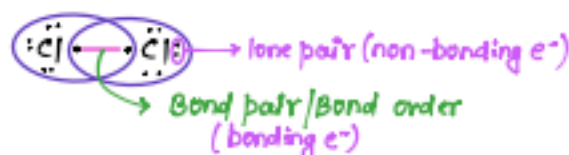
- Hydrogen bond
- van der Waal forces

② Octate Rule: atoms can combine to complete their octate

$\rightarrow$ transfer of electron:



$\rightarrow$ sharing of electron



$\rightarrow$ H & He: forms **duplet**

$\rightarrow$ For 2nd period element: central atom can form bonds according to their valency

$\rightarrow$ $\text{CH}_4, \text{O}=\text{O}, \text{NH}_3 \dots$

* Limitation:

• Hypervalent compound: ($ve > 8$) $\rightarrow$ $\text{PCl}_5, \text{SF}_6, \text{H}_2\text{SO}_4$

$\rightarrow$ use vacant 'd-orbital'

• Hypovalent compound: ($ve < 8$) $\rightarrow$ $\text{BF}_3, \text{AlCl}_3, \text{LiCl}$

• odd e^- species: $\text{NO}, \text{NO}_2, \text{ClO}_2, \text{ClO}_3$

• compound $\rightarrow$ inert gas + O/F: $\text{XeF}_2, \text{XeOF}_2, \text{KrF}_2 \dots$

* Octate rule not account shape of molecule & not explain relative stability of molecules

③ Formal charge: valid for those who follow octet Rule.

$$\text{Bond order} = \frac{\text{no. of bonding } e^-}{2} = \frac{\text{Total (Ve)} - \text{Total AVI (Ve)}}{2}$$

$\xrightarrow{8e^-/\text{atom}}$ $\xrightarrow{Ve \text{ in free state}}$ $\xrightarrow{\text{of all atom}}$

(no of bonds)

$$\text{Formal charge} = \text{Actual } Ve^- \text{ in free state of atom} - \text{Remaining } Ve^- \text{ after bonding.}$$

$$\text{Avg BO} = \frac{\text{no of Bonds}}{\text{no of surrounding atom}}$$

$$\text{Avg FC} = \frac{\text{Net charge}}{\text{no of surrounding atom}}$$

eg:

	O ₃	CO	CO ₂	N ₃ ⁻ (azide)
Bond order	$\frac{8 \times 3 - 6 \times 3}{2} = 3$	$\frac{8 \times 2 - (4 + 6)}{2} = 3$	$\frac{8 \times 2 - (4 + 6 + 6)}{2} = 4$	$\frac{8 \times 3 - (5 \times 3 + 1)}{2} = 4$
Formal charge				
Avg BO	3/2	3/1		
Avg FC	0	0		

These are not possible
b/c the formed structure should be:

- Stable by less charge distribution (or)
- Stable by symmetrical charge distribution

→ example: (oxoanions)

	SO ₄ ²⁻	PO ₄ ³⁻	NO ₂ ⁻	
Str:				• $\ddot{O} = / - \ddot{O}^-$ • $\ddot{O} \equiv / = \ddot{O}^-$ • $\ddot{O}^-$
Avg FC:	-2/4	-3/4	-1/2	
Avg BO:	6/4	5/4	3/2	

④ Covalent Bond (Bond formed by equal sharing of e⁻)

→ valency: combining capacity of atom

→ covalency: valency in covalent bond

★ * 2nd period: Max^m covalency: ④ (b/c 2d orbital not exist)

* 3rd period: can extend their valency more than 4 (b/c of 3d orbital)

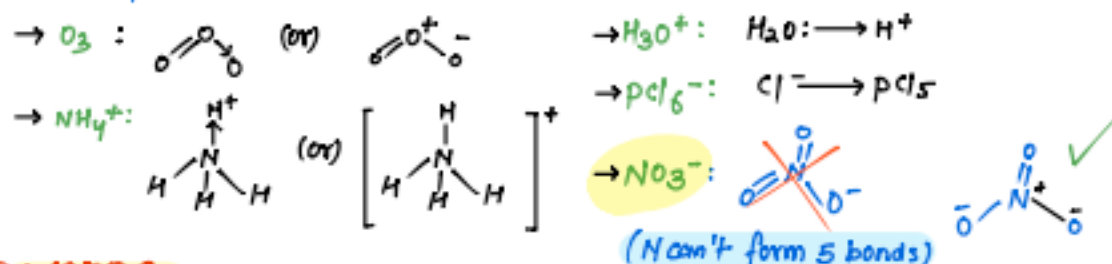
→ covalency → F = 1

→ O = 2 (but 3 if O⁺)
(+ 1 if O⁻)

⑤ Coordinate bond $\rightarrow \sigma$ -coordinate

- Formed by unequal sharing of electron
- Formed b/w e^- -donor & e^- -accepter
(Lewis base) (Lewis acid) $(A \rightarrow B) / (A^+ - B^-)$

example:



⑥ VSEPR

The main postulates of VSEPR theory are as follows:

- The shape of a molecule depends upon the number of valence shell electron pairs (bonded or nonbonded) around the central atom.
- Pairs of electrons in the valence shell repel (repel) each other since their electron clouds are negatively charged. A/R
- These pairs of electrons tend to occupy such positions in space that minimise repulsion and thus maximise distance between them.
- The valence shell is taken as a sphere with the electron pairs localising on the spherical surface at maximum distance from one another.
- A multiple bond is treated as if it is a single electron pair and the two or three electron pairs of a multiple bond are treated as a single super pair.
- Where two or more resonance structures can represent a molecule, the VSEPR model is applicable to any such structure.

* Overall Geometry depend on: σ -bond & lone pair

* Repulsive interaction: $\text{lone pair} - \text{lone pair} > \text{lone pair} - \text{bond pair} > \text{bond pair} - \text{bond pair}$
 B/c lone pair acquire more space on central atom

★ Basic



Trick: $(n + m)$

n : no of Surrounding atom

m : $\frac{(\text{ve}^- \text{ of } A) - (\text{no of SA} \times \text{valency of SA}) - (\text{charge})}{2}$

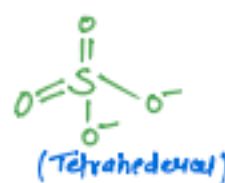
eg: SO_4^{2-}

$\rightarrow AB_4L_m$

$$m = \frac{6 - (4 \times 2) - (-2)}{2}$$

$$m = \frac{6 - 8 + 2}{2} = 0$$

$\therefore AB_4L_0$

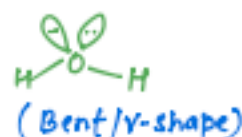


H_2O

$\bullet AB_2L_m$

$$m = \frac{6 - (2 \times 1)}{2} = 2$$

$\therefore AB_2L_2$



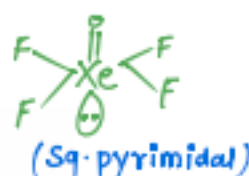
$XeOF_4$

$\bullet AB_5L_m$

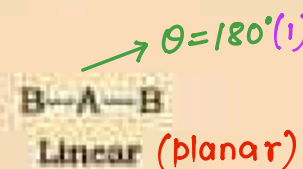
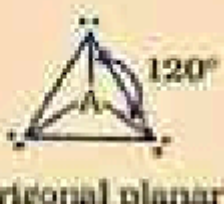
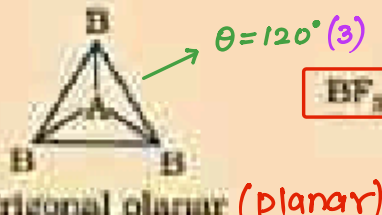
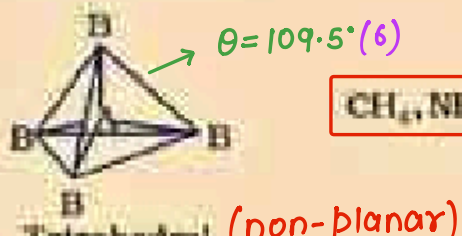
$$m = \frac{8 - (1 \times 2) - (4 \times 1)}{2}$$

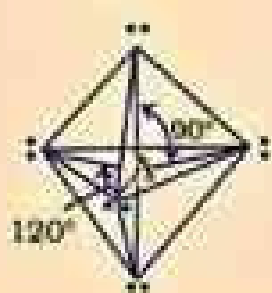
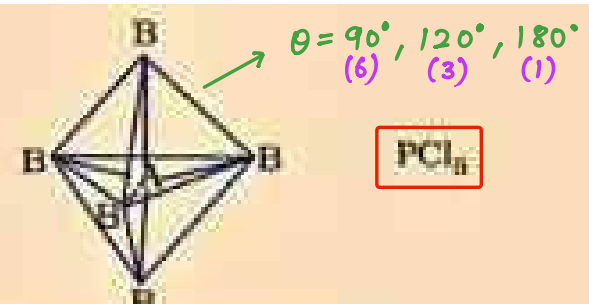
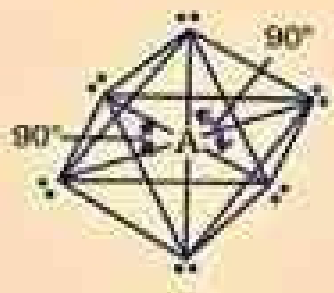
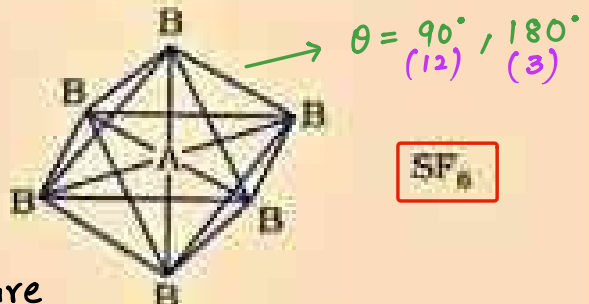
$$m = \frac{8 - 6}{2} = 1$$

$\therefore AB_5L_1$



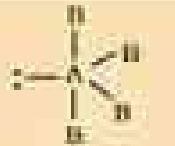
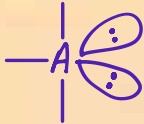
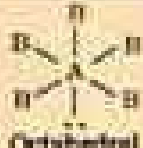
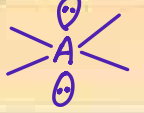
Geometry: lone pair absent

Number of electron pairs	Arrangement of electron pairs	Molecular geometry	Examples
2 AB_2Lo	 Linear	 Linear (planar)	$BeCl_2, HgCl_2$
3 AB_3Lo	 Trigonal planar	 Trigonal planar (planar)	BF_3
4 AB_4Lo	 Tetrahedral	 Tetrahedral (non-planar)	CH_4, NH_4^+

Number of electron pairs	Arrangement of electron pairs	Molecular geometry	Examples
5 AB_5Lo	 Trigonal bipyramidal	 Trigonal bipyramidal (non-planar)	PCl_5
6 AB_6Lo	 Octahedral	 Square bi-pyramidal/Octahedral (non-planar)	SF_6

Geometry: lone pair present

Molecule type	No. of bonding pairs	No. of lone pairs	Arrangement of electron pairs	Shape	Examples
AB_2E	2	1	 Trigonal planar	Bent / V-shape 	SO_2, O_3 → planar
AB_3E	3	1	 Tetrahedral	Trigonal pyramidal 	NH_3 → non-planar
AB_2E_2	2	2	 Tetrahedral	Bent / V-shape 	H_2O → planar

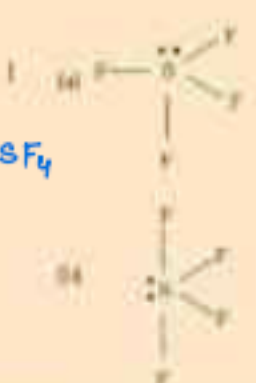
Molecule type	No. of bonding pairs	No. of lone pairs	Arrangement of electron pairs	Shape	Examples
★ AB_3E	4	1	 Trigonal bi-pyramidal	See saw 	SF_4 → non-planar
AB_2E_2	3	2	 Trigonal bi-pyramidal	T-shape 	ClF_3 → planar
AB_4E	5	1	 Octahedral	Square pyramidal 	BrF_5 → non-planar
AB_4E_2	4	2	 Octahedral	Square planar 	XeF_4 → planar

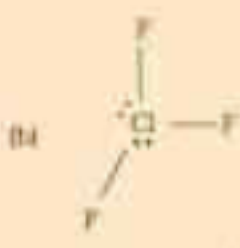
Molecule type	No. of bonding pairs	No. of lone pairs	Arrangement of electron pairs	Shape	Examples
AB_2E_3	2	3	 Trigonal bi-pyramidal	Linear 	XeF_2 planar
AB_6E_1	6	1	 pentagonal bi-pyramidal (AB_7Lo)	Pentagonal pyramidal / distorted octahedral 	XeF_6 non-planar
AB_5E_2	5	2	 pentagonal bi-pyramidal	Pentagonal planar 	XeF_5^- planar

Reason for shape

AR

Molecule type	No. of bonding pairs	No. of lone pairs	Arrangement of electrons	Shape	Reason for the shape acquired
AB_2E	4	1	 $\rightarrow SO_2$	Bent	Theoretically the shape should have been <u>triangular planar</u> but actually it is found to be <u>bent or v shaped</u> . The reason being the <u>lone pair-bond pair repulsion is much more</u> as compared to the <u>bond pair-bond pair repulsion</u> . So the angle is reduced to <u>119.5</u> from <u>120</u> . Bent Trigonal planar
AB_3E	3	1	 $\rightarrow NH_3$	Trigonal pyramidal	Had there been a <u>bp in place of lp</u> the shape would have been <u>tetrahedral</u> but <u>one lone pair is present</u> and due to the <u>repulsion between lp-lp</u> (which is more than <u>bp-bp</u> repulsion) the angle between bond pairs is reduced to <u>107</u> from <u>109.5</u> . Trigonal pyramidal tetrahedral

Molecule type	No. of bonding pairs	No. of lone pairs	Arrangement of electrons	Shape	Reason for the shape acquired
AB_2E_2	2	2	 	Bent	The shape should have been tetrahedral if there were all bp but two lp are present so the shape is <u>distorted tetrahedral</u> or <u>angular</u>. The reason is <u>lp-lp repulsion is more than lp-bp repulsion</u> which is more than bp-bp repulsion. Thus, the angle is reduced to <u>104.5</u> from <u>109.5</u>. angular tetrahedral
AB_2E	4	1	 	See saw	In (a) the lp is present at axial position so there are <u>three lp-bp repulsions at 90°</u>. In (b) the lp is in an <u>equatorial position</u>, and there are <u>two lp-bp repulsions</u>. Hence, <u>arrangement (b) is more stable</u>. The shape shown in (b) is described as a <u>distorted tetrahedron</u>, a <u>folded square</u> or a <u>see saw</u>.

Molecule type	No. of bonding pairs	No. of lone pairs	Arrangement of electrons	Shape	Reason for the shape acquired
AB_3E_2	3	2	 	T-shape	In (a) the lp are at <u>equatorial position</u> so there are <u>less lp-bp repulsions</u> as compared to others in which the lp are at <u>axial positions</u>. So structure (a) is most stable. <u>T-shaped</u>.
$\rightarrow ClF_3$			 		
			 		

⑦ Hybridisation

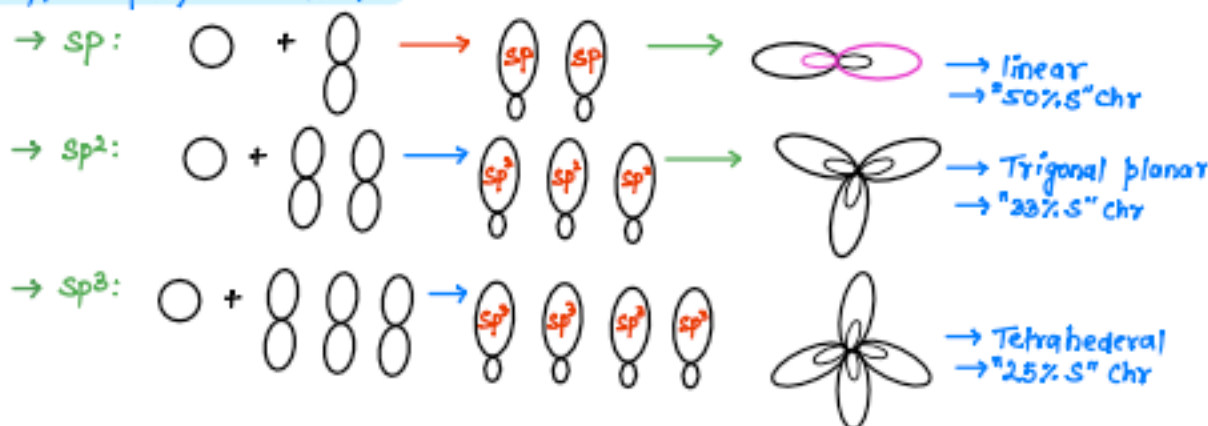
Salient features of hybridisation:

1. The number of hybrid orbitals is equal to the number of the atomic orbitals that get hybridised.
2. The hybridised orbitals are always equivalent in energy and shape.
3. The hybrid orbitals are more effective in forming stable bonds than the pure atomic orbitals.
4. These hybrid orbitals are directed in space in some preferred direction to have minimum repulsion between electron pairs and thus a stable arrangement. Therefore, the type of hybridisation indicates the geometry of the molecules.

Important conditions for hybridisation:

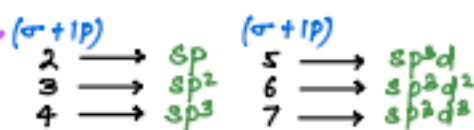
- (i) The orbitals present in the valence shell of the atom are hybridised.
- (ii) The orbitals undergoing hybridisation should have almost equal energy.
- (iii) Promotion of electron is not essential condition prior to hybridisation.
- (iv) It is not necessary that only half filled orbitals participate in hybridisation. In some cases, even filled orbitals of valence shell take part in hybridisation.

① Types of Hybridisation

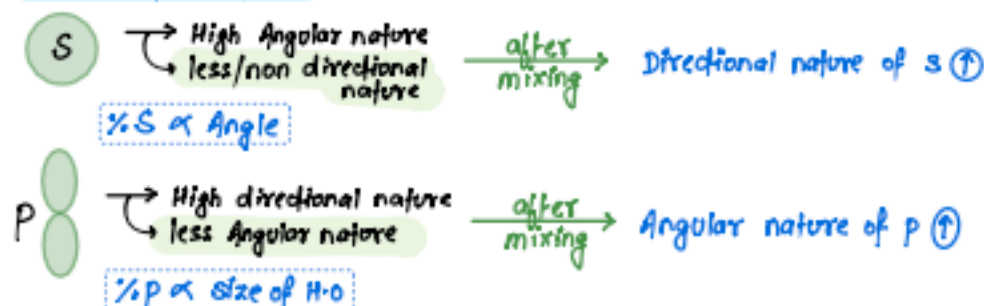


② Some imp points:

- It's orbitals Hybridised (not e^-)
- orbitals having paired e^- , single e^- or vacant orbital : all participate in Hybridisation
- During mixing of s & p orbitals: → directional nature of s ↑
→ Angular nature of p ↑
- No. of Hybrid orbitals = $\sigma \text{ bond} + \text{lone pair}$



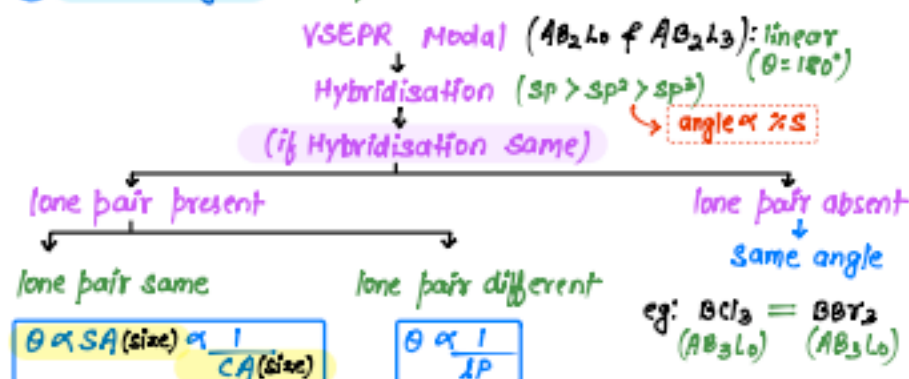
③ Nature of 's' & 'p'



⑧ Application of Hybridisation

- Bond angle
- Types of d-orbitals in Hybridisation
- Hybridisation in odd e^- species
- Hybridisation in solid state
- Existing & non-existing molecule

a) Bond angle (Trick)



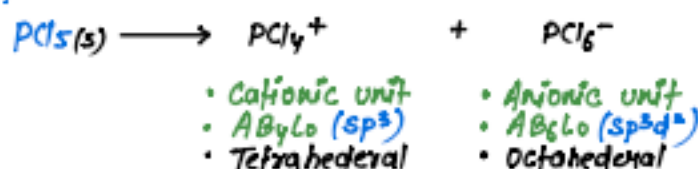
eg:

- $XeF_2 = CO_2$
 $(AB_2L_3) \quad (AB_2L_0) \rightarrow \text{linear}$
- $CH_4 > NH_3 > H_2O$
 $(AB_4L_0) \quad (AB_3L_1) \quad (AB_2L_2): sp^3$
- $NF_3 < NO_2 < NBr_3$
 $AB_3L_1 \quad (\theta \propto SA)$
- $NO_2 > PCl_3 > AsCl_3$
 $AB_3L_1 \quad (\theta \propto 1/CA)$
- PYQ • $NO_2^+ > NO_2 > NO_3^- > NO_2^-$
 $(AB_2L_0) \quad \theta = 134^\circ \quad (AB_3L_0) \quad (AB_3L_4)$
 $sp \quad \quad \quad sp^2 \quad \quad \quad sp^2$

b) Type of d-orbital use in Hybridisation

- sp^3d : dz^2 eg: $SF_6 \rightarrow AB_6L_0 (sp^3d^2)$
- sp^3d^2 : dz^2, dx^2-y^2 dz^2, dx^2-y^2
- sp^3d^3 : dz^2, dx^2-y^2, dxy

c) Hybridisation in solid state



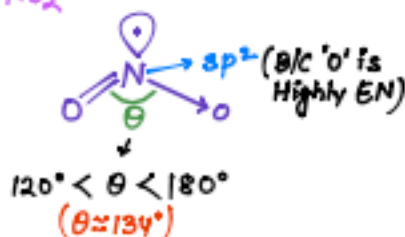
d) Hybridisation in odd e^- species

→ orbitals having odd e^- can participate in Hybridisation only when SA is Highly EN



if SA is "H" then that odd e^- does not participate in Hybridisation

eg: NO_2



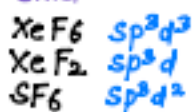
- Acidic oxide
- Cause acid rain
- paramagnetic (unpaired $e^- = 1$)
- Brown colored gas
- Dimerise under
- high pressure & Low Temp^r



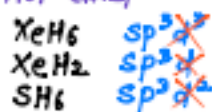
© Existing & Non-existing Molecules

→ **Non-contraction of d-orbital** → (if SA is "H" then "d-orbital" not in use)

• Exist



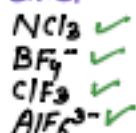
• Not-exist



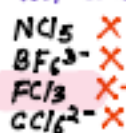
(CA required High EN SA to use d-orbital in hybridisation)

→ **Improper covalency**

• Exist



• Not exist



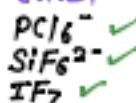
→ 2nd period: Max^m covalency = 4

→ 3rd period: Extend their covalency more than 4.

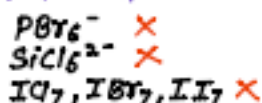
→ F → covalency: 1 (only)

→ **Steric Hindrance**

• Exist



• Not exist

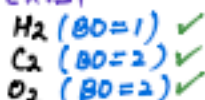


→ if SA size large

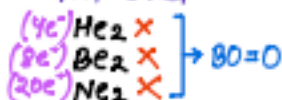
↓
 steric hindrance ✓

→ **Bond order**

• Exist



• Not exist



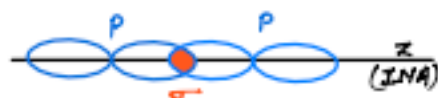
→ BO=0 : Not exist

→ BO=Fractional ✓: Exist

⑨ Overlapping (Mechanism of covalent bond formation)

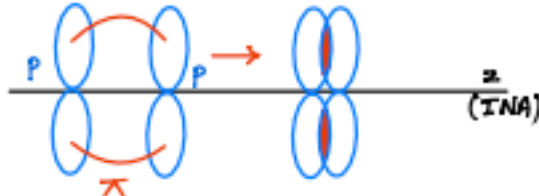
Coaxial/Head-on overlapping

- σ bond formation
- along internuclear axis



Collateral / sideways overlapping

- π bond formation (perpendicular to σ bond)
- perpendicular to internuclear axis



⇒ **Types of overlapping**

① S-S: overlap b/w two Half filled S-orbital along INA



→ σ bond

② S-P: Overlap b/w Half filled S-orbital + Half filled P-orbital



→ σ bond

③ p-p: overlap b/w two Half filled P-orbital

coaxially (along INA)

collaterally ($\perp$ to INA)



→ σ bond



→ π bond

(d) positive overlap

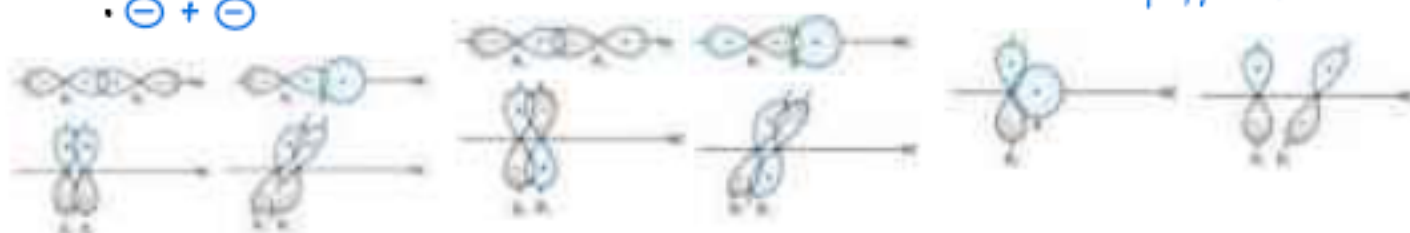
- $\oplus + \oplus$ Bond ✓
- $\ominus + \ominus$

Negative overlap

- $\oplus + \ominus$ Bond ✗

Zero overlap

- different orientation direction of approach



(10) Bond Energy / Bond Strength / Bond dissociation energy

Bond strength $\propto \frac{1}{\text{size}}$ $\propto$ directional nature (if size same)

$d > p > s$

$\Rightarrow$ BS: Co-axial $>$ Co-lateral

BS: $2p\pi - 2p\pi > 3d\pi - 3d\pi > 3p\pi - 3d\pi > 3p\pi - 3p\pi$ (weakest, Not exist)

- $N_2 (N \equiv N)$: $2p\pi - 2p\pi$ ✓
- $P_2 (P \equiv P)$: $3p\pi - 3p\pi$ ✗

$\therefore P_2$ do not exist

Imp lone pair - lone pair repulsion Results $\rightarrow$ exceptions in 2nd period element Bond strength (FON)

BS:

- $N-N < P-P$
- $F_2 < Cl_2 > Br_2 > I_2$
- $O-O < S-S$

$Cl_2 > Br_2 > F_2 > I_2$ (BS/OE/BDE)

(11) Dipole moment (μ) $\rightarrow$ net polarity of molecules

- Vector quantity • $\mu = q \times d$
- Direction: towards more EN element
- Direction of lone pair: Always away (outwards) from central atom
- if $\mu_{net} = 0$ (Non-polar) { for symmetrical str: $\mu = 0$ }
- if $\mu_{net} \neq 0$ (polar) $\rightarrow AB_2Lo$ to AB_7Lo (SA must same)

$\mu \uparrow$

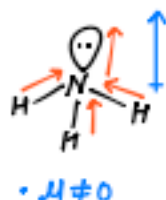
eg: PCl_5 (AB_5Lo)



XeF_2 (AB_2Lo)



NH_3



NF_3



$\mu_{NH_3} > \mu_{NF_3}$

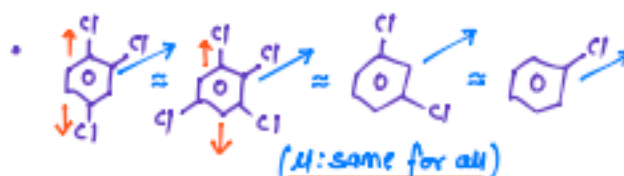
• $HF > HCl > HBr > HI$

• $H_2O > HF > NH_3 > PH_3$ (next)

• $H_2O > H_2S$

* $CH_3-Cl > CH_3-F > CH_3-Br > CH_3-I$

• $CH_3Cl > CH_2Cl_2 > CHCl_3 > CCl_4 \rightarrow (\mu \propto 1/Cl)$



(μ : same for all)

⑫ Ionic Bond / Electrovalent bond

- Complete transfer of e^- ($F \propto \frac{q_1 q_2}{r^2}$)
- Generally formed b/w Metals & non-metals (electropositive) (electronegative)
- Generally solid compounds (strong electrostatic force of attraction)
- Brittle in nature (at room temp)
- do not show isomerism (as directional nature absent)
- They are polar
- Have High MP & High BP (strong electrostatic force of attraction)

⑬ Electrical conductivity: (ions/mobility)

Ionic compound covalent compound

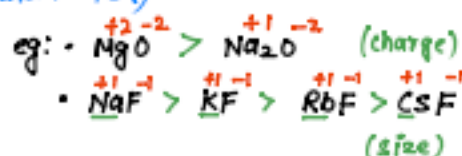
Solid state:	X	X
✓ molten state:	✓	X
Aqueous state:	✓	✓

* Metals are always conductor b/c presence of free e^- .

⑭ Lattice energy (ionic character)

$$LE \propto \frac{q_1 q_2}{r} \propto \frac{1}{\text{size}}$$

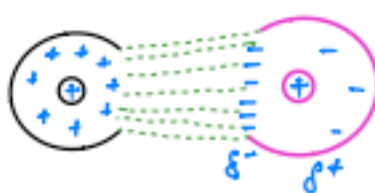
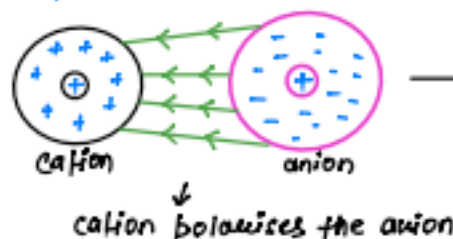
1st priority



The Lattice Enthalpy of an ionic solid is defined as the energy required to completely separate one mole of a solid ionic compound into gaseous constituent ions.

⑮ Fajan's rule: partial covalent character in ionic compounds

→ polarisation:



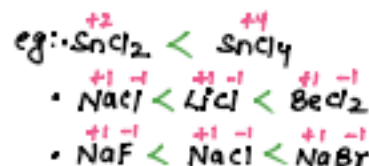
• distortion of e^- cloud of anion
 ↓
 polarisation of anion

∴ covalent character ↑

→ covalent character

$$\phi \propto q^+ \propto q^- \propto \frac{1}{\text{anionic size}} \propto \frac{1}{\text{cationic size}} \propto \frac{1}{\text{melting point}}$$

* (covalent chr $\propto$ polarising power)



Imp

The partial covalent character of ionic bonds was discussed by Fajans in terms of the following rules:

- The smaller the size of the cation and the larger the size of the anion, the greater the covalent character of an ionic bond.
- The greater the charge on the cation, the greater the covalent character of the ionic bond.
- For cations of the same size and charge, the one with electronic configuration $ns^0 np^0$, typical of transition metals, is more polarising than the one with a noble gas configuration, $ns^2 np^6$, typical of alkali and alkaline earth metal cations.

★ The cation polarises the anion, pulling the electronic charge toward itself and thereby increasing the electronic charge between the two. This is precisely what happens in a covalent bond, i.e., buildup of electron charge density between the nuclei. The polarising power of the cation, the polarisability of the anion and the extent of distortion (polarisation) of anion are the factors, which determine the per cent covalent character of the ionic bond.

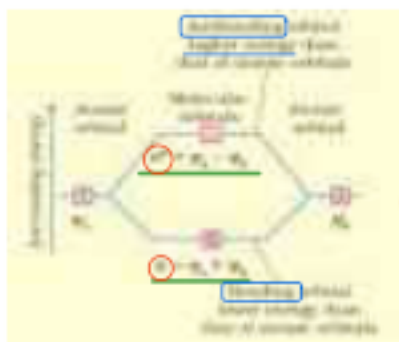
14) Molecular orbital Theory

Molecular orbital (MO) theory was developed by F. Hund and R.S. Mulliken in 1932. The salient features of this theory are:

- The electrons in a molecule are present in the various molecular orbitals as the electrons of atoms are present in the various atomic orbitals.
- The atomic orbitals of comparable energies and proper symmetry combine to form molecular orbitals.
- While an electron in an atomic orbital is influenced by one nucleus, in a molecular orbital it is influenced by two or more nuclei depending upon the number of atoms in the molecule. Thus, an atomic orbital is mono-centric while a molecular orbital is poly-centric.
- The number of molecular orbital formed is equal to the number of combining atomic orbitals. When two atomic orbitals combine, two molecular orbitals are formed. One is known as bonding molecular orbital while the other is called antibonding molecular orbital.
- The bonding molecular orbital has lower energy and hence greater stability than the corresponding antibonding molecular orbital.
- Just as the electron probability distribution around a nucleus in an atom is given by an atomic orbital, the electron probability distribution around a group of nuclei in a molecule is given by a molecular orbital.
- The molecular orbitals like atomic orbitals are filled in accordance with the Aufbau principle obeying the Pauli's exclusion principle and the Hund's rule.

(a) LACO: linear combination of atomic orbitals

→ the atomic orbitals can be expressed by wave function (ψ) → represent amplitude of electron wave.



ABMO: • In formation of ABMO, the electron wave cancel each other due to destructive interference.

• The electron placed in ABMO destabilise the molecule b/c of mutual repulsion of electrons is more than attraction.

BMO: • In formation of BMO, the electron wave reinforce each other due to constructive interference.

• The electron placed in BMO, hold the nuclei & stabilise the molecule. ∴ BMO always possesses lower energy.

★ It may be noted that the energy of the antibonding orbital is raised above the energy of the parent atomic orbitals that have combined and the energy of the bonding orbital has been lowered than the parent orbitals. The total energy of two molecular orbitals, however, remains the same as that of two original atomic orbitals.

(b) Conditions for combination of Atomic orbitals

→ Combining orbitals must:

- have same (nearly same) energy.
- have same symmetry about molecular axis.
- overlap to maximum extent.

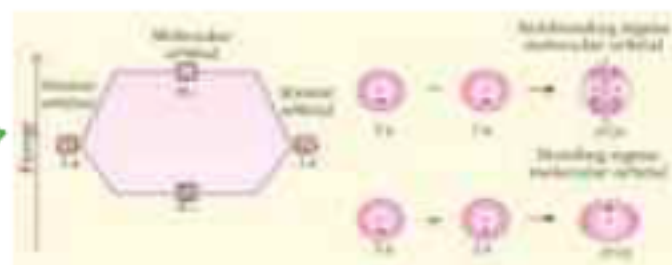
(c) Types of molecular orbitals:

Molecular orbitals of diatomic molecule → σ, π, δ

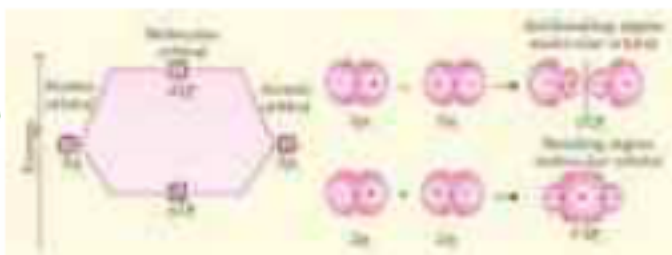
• σ (BMO/ABMO) → Symmetrical

• π (BMO/ABMO) → unsymmetrical

(i) the linear combination of $1s$ orbitals centered on two nuclei produces two molecular orbitals which are symmetrical around the bond-axis. Such molecular orbitals are of the σ type and are designated as $\sigma 1s$ and $\sigma^* 1s$.



(ii) If internuclear axis is taken to be in the x -direction, it can be seen that a linear combination of $2p_x$ orbitals of two atoms also produces two sigma molecular orbitals designated as $\sigma 2p_x$ and $\sigma^* 2p_x$.



(iii) Molecular orbitals obtained from $2p_x$ and $2p_y$ orbitals are not symmetrical around the bond axis because of the presence of positive lobes above and negative lobes below the molecular plane. Such molecular orbitals, are labelled as π and π^*

(BMO)

- π bonding MO has larger electron density above and below the inter-nuclear axis
- π^* antibonding MO has a node between the nuclei (ABMO)



④ Energy order:

- up to N_2 : $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \pi 2p_x < \pi 2p_y < \sigma 2p_z < \pi^* 2p_x < \pi^* 2p_y < \sigma^* 2p_z$
(2121) $\rightarrow O_2, Ca, N_2$
- After N_2 : $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \pi 2p_x < \pi 2p_y < \pi^* 2p_x < \pi^* 2p_y < \sigma^* 2p_z$
(1221) $\rightarrow O_2, F_2$

⑤ Bond order

$$BO = \frac{N_b - N_a}{2}$$

Annotations: N_b is BMO e^- , N_a is ABMO e^-

$N_b > N_a$: molecule stable
 $N_b < N_a$ (or) $N_b = N_a$: molecule unstable

$$BO \propto BS \propto \frac{1}{BL}$$

⑥ Bond order trick

e^- :	1	2	3	4	5	6	7	8
BO:	0.5	1	0.5	0	0.5	1	0.5	0

e^- in $\rightarrow$ BMO: BO inc by 0.5
 $\rightarrow$ ABMO: BO dec by 0.5

$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s$

WIP

e^- :	8	9	10	11	12	13	14	15	16	17	18	19	20
BO:	0	0.5	1	1.5	2	2.5	3	2.5	2	1.5	1	0.5	0

Annotation: 14 is N_2

eg: $O_2^- (17e^-) \rightarrow (1221)$
 $BO = 1.5$

For further detail: $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \pi 2p_x < \pi 2p_y < \pi^* 2p_x < \pi^* 2p_y < \sigma^* 2p_z$

Annotations: $8e^-$ in σ orbitals, $(17-8=9)$ in π and π^* orbitals.

e^- in BMO = 6 (2 in σ BMO, 4 in π BMO)
 e^- in ABMO = 3 (all in π^* ABMO)

$$BO = \frac{6-3}{2} = \frac{3}{2} = 1.5$$

⑧ Magnetic nature

- e^- : even number $\rightarrow$ Diamagnetic
 $\rightarrow$ except: $B_2, O_2, S_2 \rightarrow$ paramagnetic
 (10e) (16e) (32e)
- e^- : odd number $\rightarrow$ paramagnetic
 $\rightarrow$ at least 1 unpaired e^-

* magnetic moment $= \sqrt{n(n+2)}$
 $\rightarrow$ MM $\propto n \propto$ paramagnetic nature
 $\rightarrow n=2 \therefore MM=2$

⑮ Hydrogen bond (-----)

→ special case of dipole-dipole interaction



Inter-molecular



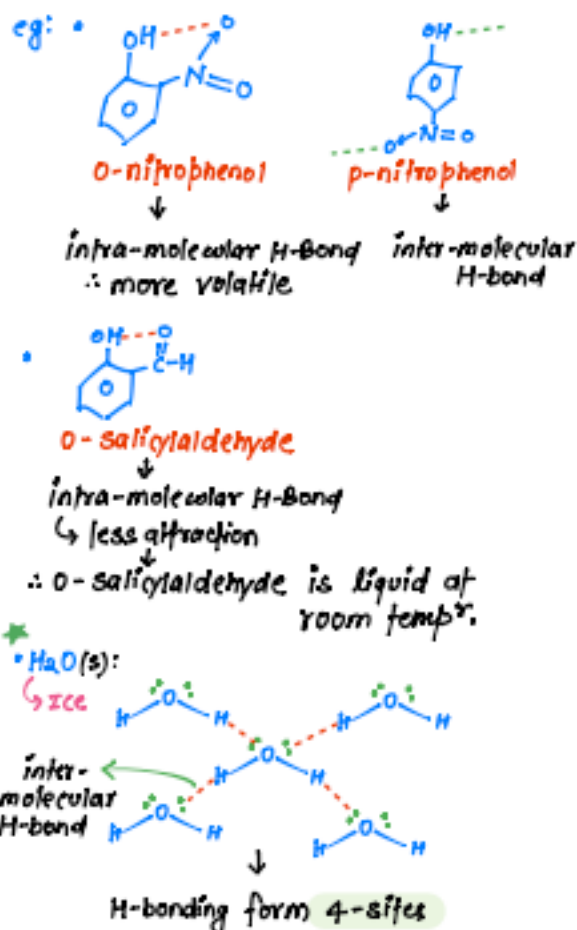
- b/w two molecule
- Chelation (ring) may occur
- More association
 - High MP, BP &
 - less volatile
 - low vapour pressure
 - More viscosity

Intra-molecular



- within a molecule
- Chelation (ring) must occur
- less association
 - low MP, BP
 - More volatile
 - High Vapour pressure
 - less viscosity

The magnitude of H-bonding depends on the physical state of the compound. It is maximum in the solid state and minimum in the gaseous state. Thus, the hydrogen bonds have strong influence on the structure and properties of the compounds.



⑮ Dipole based forces

Ions involved

- Ion-dipole (b/w ion & polar)
eg: Li^+ , H_2O
- Ion-induced dipole (b/w ion & non polar)
eg: Na^+ , Br_2

Ions not involved

(vanderwall forces/IMF)

- Dipole - Dipole (b/w polar)
eg: H_2O , H_2O (H-bond) → special case of DP-DP
 • HF , HF
- Dipole-induced dipole (b/w polar & non-polar)
eg: H_2O , Xe
 • H_2O , Ne
- London dispersion forces (b/w non-polar)
eg: Br_2 , Br_2
 • H_2 , H_2

P BLOCK

Inert pair effect

- $T \rightarrow B$: • lower O.S stability $\uparrow$
• Higher O.S stability $\downarrow$

② : Due to poor d & f shielding
 $\therefore Z_{eff} \uparrow$

Group 13

① * Boron compound

- Borax : $Na_2B_4O_7 \cdot 10H_2O$
- Orthoboric acid : H_3BO_3
- Kernite : $Na_2B_4O_7 \cdot 4H_2O$
- Borax bead : $NaBO_2$
- Diborane : B_2H_6
- Borazine : $B_3N_3H_6$ (inorganic benzene)

* Aluminium compound

- Bauxite : $Al_2O_3 \cdot 2H_2O$
- Cryolite : Na_3AlF_6

② Atomic Radius :

$$\Rightarrow B < Al > Ga < In < Tl$$

$$\hookrightarrow (B < Ga < Al < In < Tl)$$

③ IE: $B > Tl > Ga > Al > In$
(due to poor d & f shielding)
 $\rightarrow$ T.C & L.C

④ EN:

$$B > Al < Ga < In < Tl : (Al < Ga < In < Tl < B)$$

(1st dec, then Inc)

⑤ Melting point

$$\hookrightarrow B > Al > Tl > In > Ga$$

① Physical properties

① Boron

- non metallic, Hard, black, crystalline
- Have very high MP $\rightarrow$
(b/c it has strong crystalline lattice)

② Gallium

- Low MP (303K), High BP (2676K)
 $\downarrow$
exist in liq. state $\therefore$ it measures high Temp

② Chemical properties

* inert pair effect:

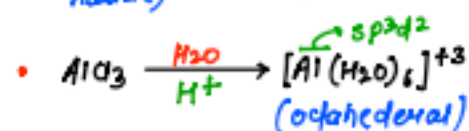
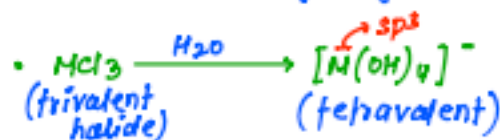
- Stability of +1 O.S $\rightarrow Al < Ga < In < Tl$
($T \rightarrow B: \uparrow$)
($Al > Ga > In > Tl$)

- predominant on Tl : ($Tl^{+1} > Tl^{+3}$)

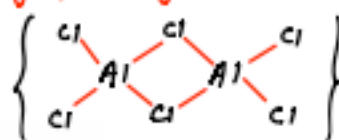
* BF_3

$\rightarrow sp^2$, electron deficient, Lewis acid
(Lewis acid $\propto \frac{1}{Size(CA)}$ $\propto$ size(SE))
tendency

* In trivalent halides, most of compounds are covalent & Hydrolysed in water



* $AlCl_3$ achieve stability by forming dimer $\rightarrow Al_2Cl_6$



③ Reactivity

Air (O_2)

- Crystalline Boron X
- Amorphous (B/Al) $\rightarrow B_2O_3 / Al_2O_3$

B_2O_3 : acidic
 Al_2O_3 / Ga_2O_3 : Ampho
 In_2O_3 / Tl_2O_3 : Basic

Acids/Alkalies

- Boron X
- Al $\xrightarrow[\text{aq. alkali}]{\text{dil. HCl}}$ $H_2 \uparrow$
- Al $\xrightarrow{C \cdot HNO_3}$ protective oxide layer

Halogen

- E $\xrightarrow{X}$ EX_3 (trihalide)
- TlI_3 X (inert pair)
- $TlI > TlI_3$

④ Anomalous prop. of Boron

- $EX_3 \rightarrow$ covalent & Hydrolysed in water (monomeric trihalide)
- electron deficient, Lewis acid

★ Except: Boron

can exist in aq. medium

* Boron Max^m covalency = 4 & absence of 'd orbital'.

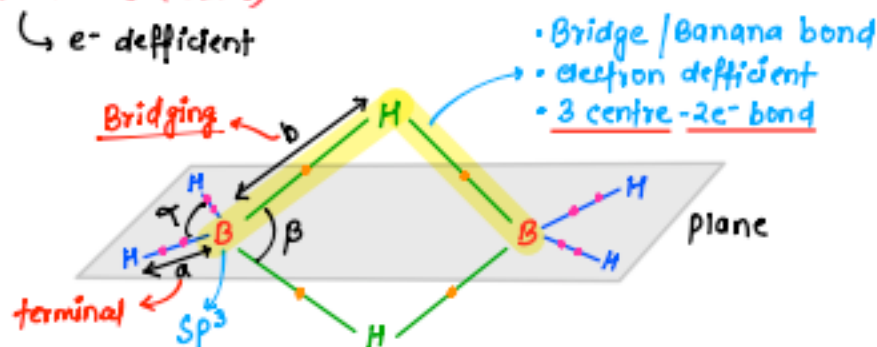
* $\therefore BF_6^{3-}$ X • AlF_6^{3-} ✓

⑤ Uses

- Boron: • Bullet proof vest
• Aircraft material
- Borax & boric acid: • Heat resistant glass
• Glass-wool
• Fibre glass
- Aluminium: • High conductivity
• Form alloy with (Mg, Si, Mn, Cu, Zn)

⑥ Diborane (B_2H_6)

$\hookrightarrow e^-$ deficient



- Bond length: $b > a$
- Bond strength: $b > a$
- angle: $\alpha > \beta$

- Have 4 terminal B-H bond ($2c-2e^-$), (same plane)
- Have 2 bridging B-H-B bond ($3c-2e^-$)
- Max^m '6 atom' are in same plane (2B, 4H)

* Abundancy in earth crust: • element: O > Si > Al > Fe > Ca
 • Metals: Al > Fe > Ca

Group 14

- $\text{CO} + \text{H}_2$: water/synthetic gas
- $\text{CO} + \text{N}_2$: producer gas

• C/Si: Non-Metal
• Ge/Sn: Metalloid
• Pb: Metal

(a) IE: $\text{C} > \text{Si} > \text{Ge} > \text{Sn} < \text{Pb}$

($\text{C} > \text{Si} > \text{Ge} > \text{Pb} > \text{Sn}$)

same for MP

(b) EN: $\text{C} > \text{Si} = \text{Ge} = \text{Sn} \approx \text{Pb}$

2.5 1.8 1.8 1.8 1.9

(from Si to Pb: EN almost same)

① Chemical properties

• Inert pair: Stability of +2 o.s. ↑

$\text{Ge}^{+2} < \text{Sn}^{+2} < \text{Pb}^{+2}$

($\text{C}^{+4} > \text{Si}^{+4} > \text{Ge}^{+4} > \text{Sn}^{+4} > \text{Pb}^{+4}$)

st:

• $\text{Pb}^{+2} > \text{Pb}^{+4}$

(OA)

• $\text{Sn}^{+2} < \text{Sn}^{+4}$

(RA)

② Reactivity:

O_2

$\text{M} \xrightarrow{\text{O}_2} \text{MO}/\text{MO}_2$

* Acidity $\propto$ HOS ↑

MO_2 (Dioxides)

- $\text{CO}_2/\text{SiO}_2/\text{GeO}_2$: Acidic
- $\text{SnO}_2/\text{PbO}_2$: Ampho

MO (Monoxide)

- CO : Neutral
- GeO : Acidic (SiO)
- SnO/PbO : Ampho

water

• C/Si/Ge: X

• $\text{Sn} \rightarrow \text{SnO}_2 + \text{H}_2$

Halogen

* $\text{M} \rightarrow \text{MX}_2/\text{MX}_4$

• MX_4 (covalent)

(but, $\text{SnF}_4/\text{PbF}_4$: ionic)

• In Halogen members MX_2 is stable

st: $\rightarrow \text{PbX}_2 > \text{PbX}_4$ (IAE)

st: $\rightarrow (\text{GeX}_2 < \text{GeX}_4)$

• $\text{MX}_4 \xrightarrow{\text{H}_2\text{O}} \text{M(OH)}_4$

↳ M: X, Si ✓ ($\text{CCl}_4 \rightarrow \text{X}$)

③ Anomalous prop. of carbon

• C: $\text{p}\pi\text{-p}\pi \rightarrow (\text{C}=\text{C}, \text{C}\equiv\text{C}, \text{C}=\text{O}, \text{C}=\text{S}, \text{C}\equiv\text{N})$

(but Halogen members do not form $\text{p}\pi\text{-p}\pi$: B/c of large size)

• C $\rightarrow$ catenation ✓ (linking with each other)

↳ B/c C-C bond is strong

• T $\rightarrow$ B : catenation ↓

∴ $\text{C} \gg \text{Si} > \text{Ge} \approx \text{Sn} > \text{Pb}$

* CCl_4 : • electron precise
• C $\rightarrow$ maxm covalency: 4

* Exist: SiF_6^{2-} , GeCl_6^{2-} , Sn(OH)_6^{2-} , PbF_4

~~SiCl_6^{2-} , PbI_4~~

• large size of chlorine

• poor (Cl-Si) interaction

* ultrapure • transistor,
 Ge & $\text{Si} \rightarrow$ • semiconductor

④ Allotropes of carbon → carbon show allotropic form due to catenation & p π -p π

Diamond

- Crystalline lattice
- sp³ (3D)
- tetrahedral
- directional covalent bond
- Hardest substance
($\therefore$ used in sharpening)
- $(\Delta H_f^\circ)_{\text{diamond}} = 1.9 \text{ KJ/mol}$

Graphite

- layered structure
- sp² (2D)
- Planar Hexagonal
- Vanderwall forces
- Delocalisation of e⁻
 $\therefore$ conduct electricity
- Soft & slippery
 $\therefore$ used as lubricant
- thermodynamically most stable
- $(\Delta H_f^\circ)_{\text{graphite}} = 0$

Fullerene (C₆₀)

- Graphite $\xrightarrow[\text{He/Ar}]{\Delta}$ C₆₀
- sp²: Soccer ball like

Buckminster fullerene

- 6 member ring = 20
- 5 member ring = 12

→ C₆₀ is only pure form of carbon

↳ B/c of smooth structure
& absence of dangling bond

- $(\Delta H_f^\circ)_{\text{C}_{60}} = 38.1 \text{ KJ/mol}$

COORDINATION COMPOUNDS

① Double salt

Complete ionize

Salt lose their identity in aq. soln

Individual ion retain their identity

(∴ each ion give +ve test)

- eg: • $KCl \cdot MgCl_2 \cdot 6H_2O \rightarrow$ carnalite
 • $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O \rightarrow$ Mohr's salt
 • $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O \rightarrow$ potash alum

Complex salt

• Not complete ionize

• Salt retain their identity in aq. soln

• Individual ion loose identity

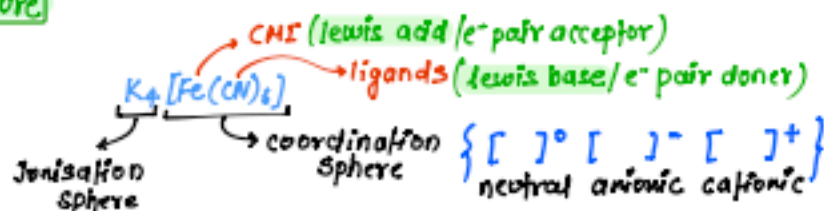
(∴ each ion don't give +ve test)

- eg: • $K_4[Fe(CN)_6]$

② Condition for coordination compounds

→ • Generally d-block metals → • Small size • High ϕ nature • Vacant orbitals

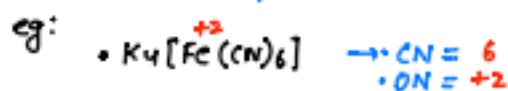
Structure



③ Some important terms

(a) **Coordination number:** no. of ligand $\times$ denticity

(b) **Oxidation number of central atom:** charge present on CMI



(c) **Homoleptic complex**

v/s

Heteroleptic complex

→ one kind of ligands

- $K_2[Al(C_2O_4)_3]$
- $[Co(NH_3)_6]^{3+}$

→ more than 1 kind of ligands

- $[Co(NH_3)_4Cl_2]^+$
- $[Co(NH_3)_4(H_2O)Cl]Cl_2$

(d) **Coordination polyhedron:** arrangement of ligand around CMI.



(e) **Ligand:** molecule/ion donate e^- pair to CMI

Lewis base → **denticity (d)** → no. of e^- pair can be donated by ligand.

④ classification of ligands

* Charge

① Anionic

• monodentate (d=1)

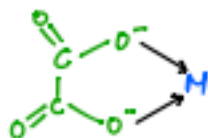
- SO_4^{2-} : sulphato
- CO_3^{2-} : carbonato
- Cl^- : Chlorido
- Br^- : Bromido
- OH^- : Hydroxido
- F^- : Fluorido
- NH_2^- : Amido
- NH^{2-} : Imido
- N_3^- : Azido
- N^{3-} : Nitrido
- O_2^- : Superoxido
- O^{2-} : Oxido
- O_2^{2-} : peroxido

have more than one donor atom.

Ambidentate (d=1)

- CN^- : Cyanido
- NC^- : Isocyanido
- NO_2^- : Nitrito-N
- ONO^- : Nitrito-O
- NCO^- : cyanato-N
- OCN^- : cyanato-O
- SCN^- : Thiocyanato-S
- NCS^- : Thiocyanato-N

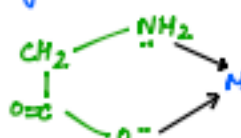
• $\text{C}_2\text{O}_4^{2-}$: oxalato (ox)



- chelation / Ring
- d=2
- donor atom = 2 'O' (Symm)

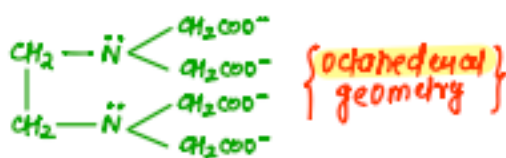
stable ↑

• Gly^- : Glycinate



- Chelation
- d=2
- donor atom: N & O (unsymm)

• EDTA^{4-} (ethylene diamine tetra acetate)



- chelation / Ring (5 member, 5 ring)
- Non-coplanar

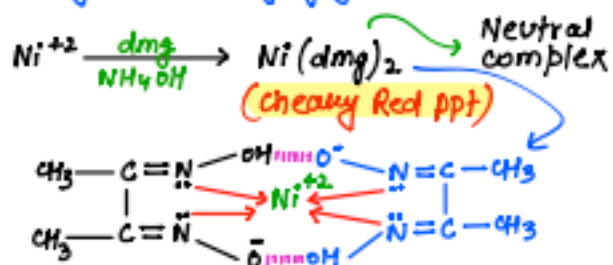
- d=6 (Hexadentate)

- Donor atom: 2 'N' & 4 'O'

- optically active ✓

- CN=6

• dmg^- (di-methyl glyoximate)



- chelation ✓ (4 Ring)
- d=2
- CN=4, dsp², sq-planar

- donor atom = 'N'
- optically inactive (POS ✓)

- Dimerisation ✓
- Symm. H-bonding

② Neutral

- H_2O : aqua
- NH_3 : Amino
- CO : Carbonyl
- NO : Nitrosyl
- PH_3 : phosphine
- PY : pyridine

(d=1)



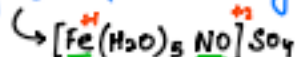
Chelation ✓

- en: $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ (d=2) ethylene-diamine
- dien: $\text{NH}(\text{CH}_2\text{CH}_2\text{NH}_2)_2$ (d=3) diethylene-triamine
- tren: $\text{N}(\text{CH}_2\text{CH}_2\text{NH}_2)_3$ (d=4) tris(2-aminoethyl)amine
- trien: $\text{NH}_2\text{CH}_2\text{CH}_2\text{NHCH}_2\text{CH}_2\text{CH}_2\text{NHCH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ (d=4) triethylene-tetraamine

④ cationic

- NO^+ : Nitrosonium (when NO with Fe) NO^+
- $\text{NH}_2\text{-NH}_2^+$: Hydrazenium

eg: BRC: (Brown ring complex)



: pentaqua nitrosonium iron (I) sulphate

⑤ IUPAC

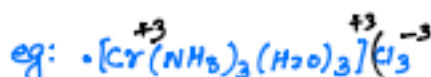
Rule: ① Name of cation 1st.

② For $[]^+ / []^0$: Number of ligand $\rightarrow$ Name of ligand $\rightarrow$ Name of Metal $\rightarrow$ o.s of Metal (Roman)

③ $[]^-$: 'ate' in anionic complex

(Bi, tri) (Alphabetically) (ate in suffix if $[]^-$)

otherwise write normally



triammine triaqua chromium(III) chloride



tris (ethylenediamine) cobalt(III) sulphate

* (if di, tri, tetra already present in name of ligand, then use bis, tris, tetrakis for number of ligand.)



potassium trioxalatoaluminate (III)



diammine silver(I) dicyanido argentate(I)

⑥ Bonding in complex

a) Werner's theory

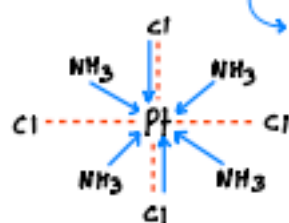
$\rightarrow$ CMI have 2 types of valencies

Primary valency

- equal to ON of CMI
- Satisfied by anion
- Ionisable (may)
- non-directional
- $---$
- Geometry X

Secondary valency

- equal to CN of CMI
- Satisfied by Neutral & anion
- NON-Ionisable
- directional (covalent)
- $---$
- Geometry $\checkmark$



ionisable (बाह्य काल) non-ionisable

- 1st V: ON = 4 (but 2 ionisable)
- 2nd V: CN = 6

• Rxn with AgNO_3

$\therefore$ no. of AgCl produced = 2 (ionisable wale Cl^-) (ppt)

	ions
$[\text{Pd}(\text{NH}_3)_4] \text{Cl}_2$	3
$[\text{Ni}(\text{H}_2\text{O})_6] \text{Cl}_2$	3
$[\text{Pt}(\text{H}_2\text{O})_2 \text{Cl}_4]$	1
$[\text{Co}(\text{NH}_3)_4 \text{Cl}_2] \text{Cl}$	2
$[\text{Pt}(\text{NH}_3)_2 \text{Cl}_2]$	1

no. of α Molar conductivity

⑥ VBT: valence bond theory

• Geometry, hybridization $\xrightarrow{\text{via}}$ **CN**

CN	Geometry	Hybridization
2	linear	sp
3	trigonal planar	sp ²
4	Sq. planar	dsp ²
4	tetrahedral	sp ³
5	Tri-bipyramidal	sp ³ d/dsp ³
6	octahedral (Sq. bipyramidal)	sp ³ d ² /d ² sp ³

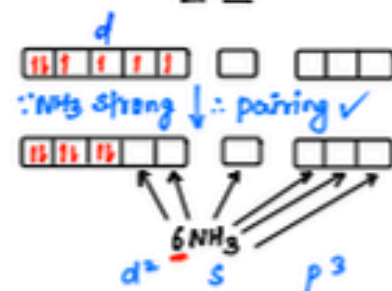
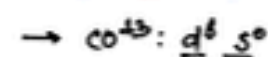
* Donor strength of Ligand

C > N > O > X (F > Cl > Br > I)	
(Strong)	(Weak)
• pairing of 'de'	• pairing ⊗
• low spin complex	• High spin complex
• diamagnetic (n=0)	• paramagnetic (n≠0)
• inner orbital complex	• outer orbital complex

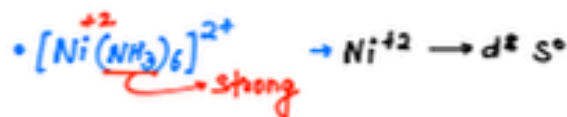
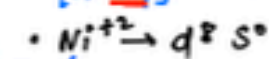
* pairing tab kro jab req ho, and jitni jarurat ho utni hi pairing kro

In general

eg:



②



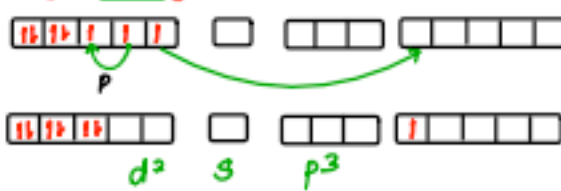
pairing possible, but Hoga nahi, kyoki after pairing Hame only 1 vacant d orbital milega, but we need '2'.



* Transfer rule (when reqd)

→ 1e⁻ inner 'd' transfer to Higher energy level

when strong ligand present. ✓

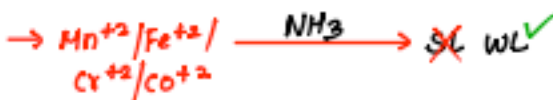
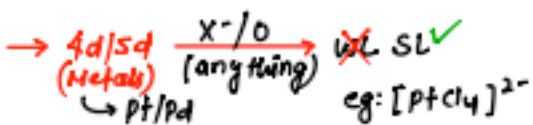
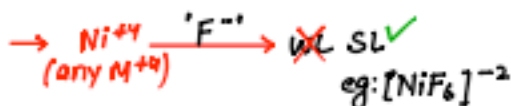
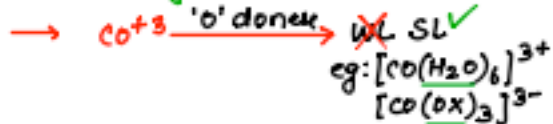


* Hybrid: $d^2 sp^3$ (octahedral)

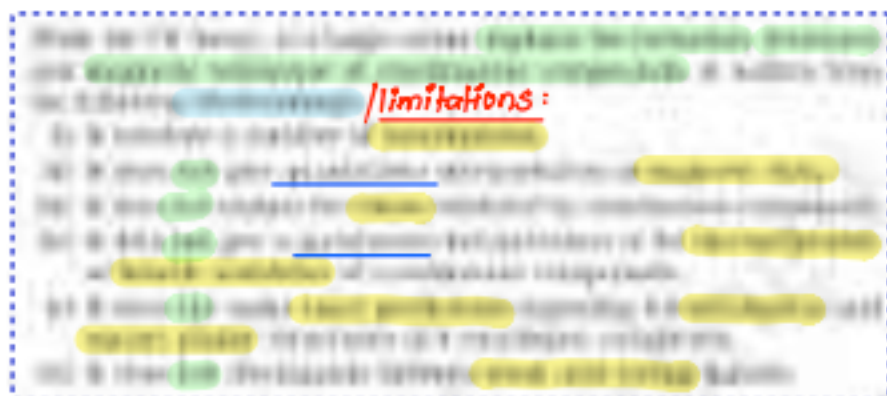
• complex: inner orbital & low spin

• Unpaired e⁻: 1 (para)

* Strength rule:



Limitations of VBT



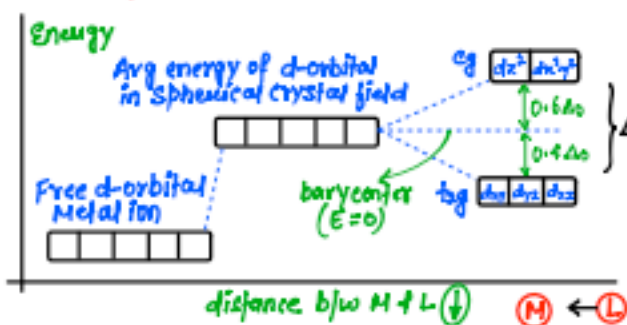
© Crystal field theory

- Based on electrostatic energy
- Metal-ligand bond: ionic

• Metal ion: +ve charge

• Ligand → anion: point charge (-ve)
→ Neutral: point dipole

★ Splitting of octahedral complex (o)



* $d_{z^2}, d_{x^2-y^2}$ (eg) → directed towards axis
→ experience more repulsion
∴ Raised its energy
eg increase its energy by (3/5) Δ_o

* d_{xy}, d_{yz}, d_{zx} (t_{2g}) → directed b/w axis
→ lowered its energy
t_{2g} decrease its energy by (2/5) Δ_o

pairing energy

Δ_o < P: weak field ligand (High spin complex)

e⁻: t_{2g} → e_g → t_{2g} → e_g

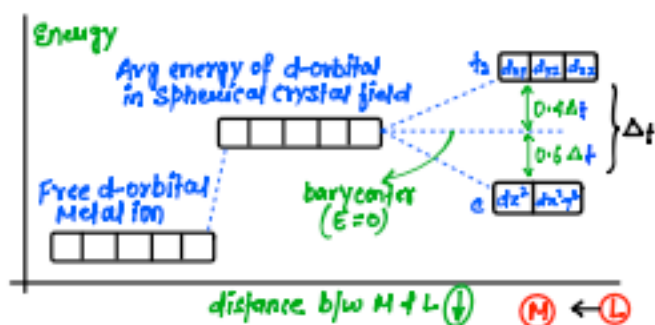


Δ_o > P: Strong field ligand (low spin complex)

e⁻: t_{2g} → t_{2g} → e_g → e_g



★ Splitting of tetrahedral complex (t) (X)



* Always weak field ligand ($\Delta_t < P$) (High spin)

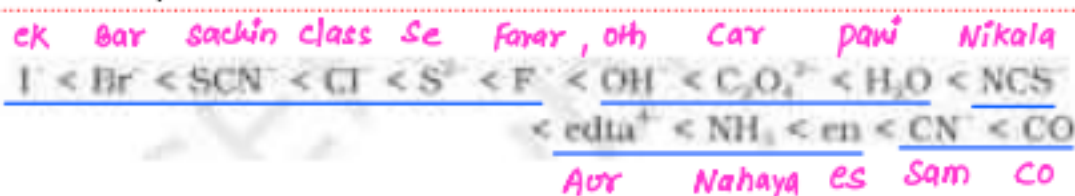


* $\Delta_t < \Delta_o$ $\Delta_t = \frac{4}{9} \Delta_o$

① Factor affecting splitting (o)

④ Field strength of ligand $\propto \Delta$

↓
Spectrochemical series (based on absorption of light)



⑥ charge on central metal ion $\propto \Delta$

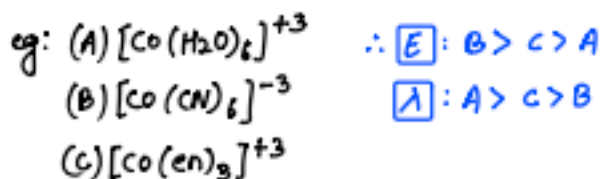


③ Colour of complex

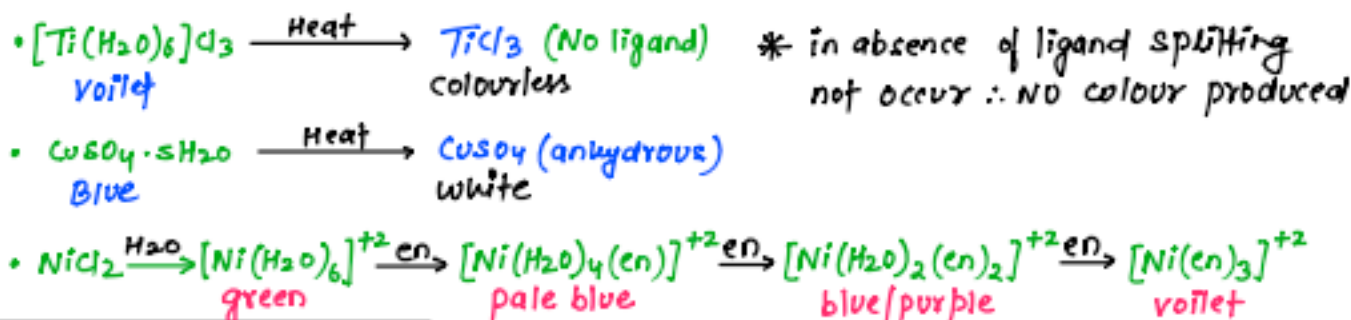
$\rightarrow$ explained by CFT & colour due to d-d transition

$\rightarrow d^0, d^{10}$: colourless, d^{1-9} : coloured

* $\Delta \propto \frac{1}{\lambda} \rightarrow$ wavelength



④ NCERT example:



colour of gem stones:

- Ruby: Cr^{+3} in $Al_2O_3 \rightarrow$ Red
- Emerald: Cr^{+3} in $Be_3Al_2Si_6O_{18}$ (Beryl) $\rightarrow$ Green

e) Stability of complex depend on

- Charge on CMI
- Field strength of ligand (SCS)
- Chelation (Ring)

Amongst the following, the most stable complex is
 (a) $[PtCl_2(OH)_2]^-$ (b) $[PtCl_2(en)]$ (c) $[PtCl_2(en)]^+$ (d) $[PtCl_2]^-$



f) Limitation of CFT

The crystal field theory is based on the assumption of electrostatic interaction between the central metal ion and the ligands. It does not take into account the covalent character of the bond. It also does not explain the magnetic properties of the complexes. It is not able to explain the color of the complexes. It is not able to explain the stability of the complexes. It is not able to explain the geometry of the complexes. It is not able to explain the reactivity of the complexes.

8) organometallic compounds (when carbon directly attach to Metals)

σ -Bonded



- $R-MgX$ (Grignard)
- R_2Zn (Frankland)
- R_2CuLi (Gilman)
- $TiCl_4 + R_3Al$ (Ziegler Natta)

π -Bonded



- Zeise's salt
- Ferrocene
- Cobaltocene
- Ruthenocene

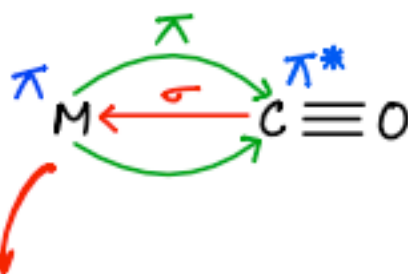
$\sigma+\pi$ Bonded



- ↓
 Metal carbonyl
 → $Ni(CO)_4$ etc.

⑨ Metal carbonyl bond / Synergic bond

Non classical ligand $\rightarrow$ CO (Neutral)



- Metal carbonyl bond posses both $(\sigma + \pi)$ character
- Synergic effect strengthen bond b/w Metal & CO. (M-C)

• (M-C) σ bond: donation of l.p of e^- from carbon to Metal (vacant orbital)

• (M-C) π bond: donation of pair of e^- from Metal to carbon (filled orbital) (Antibonding π^* orbital)

* Strength of SB

$\propto e^-$ density on metal ($M^- > M^0 > M^+$)

$\propto$ lower O.S of Metal

$\propto$ (M-C) bond order/strength

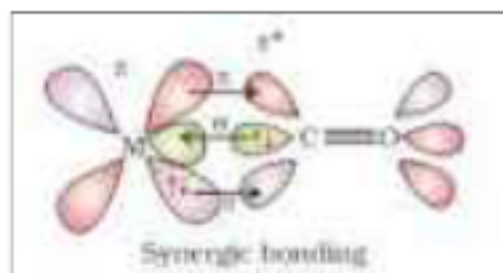
$$SB \propto (MC)BO \propto (CO)BL \quad \left(SB \propto (MC)BO \propto (CO)BL \right. \\ \left. \propto \frac{1}{(MC)BL} \propto \frac{1}{(CO)BO} \right)$$

eg: ① $[Cr(CO)_6]^0$

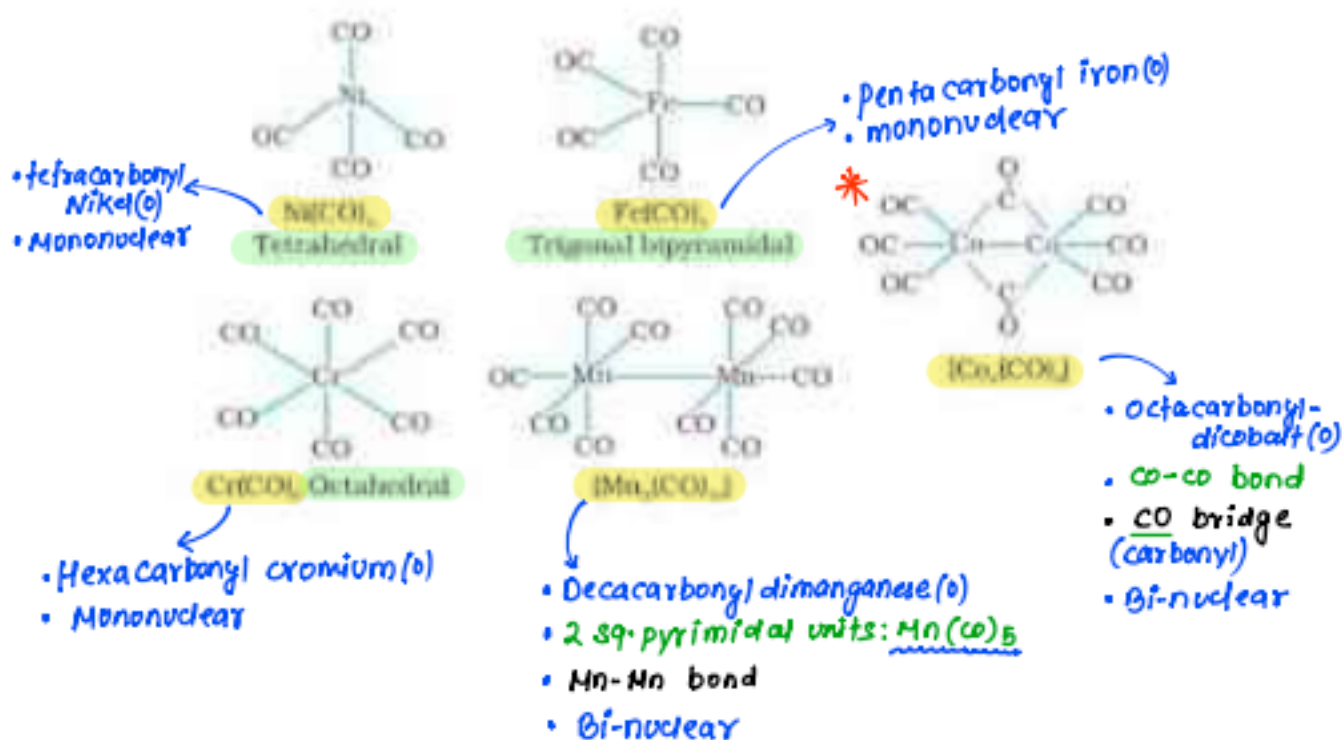
② $[V(CO)_6]^-$

③ $[Mn(CO)_5]^+$

$\rightarrow SB: B > A > C$



⑩ Structure of Metal carbonyl (all Homolyptic, all Metal in zero O.S)



⑪ Isomerism

① Structural isomerism

→ **Ionisation isomerism**: counter ion is itself ligand & have power to displace ligand

eg: $[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Br}$, $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$

→ **Linkage isomerism**: have ambidentate ligand

eg: $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$, $[\text{Co}(\text{NH}_3)_5(\text{ONO})]\text{Cl}_2$

(no of linkage isomer = 2n) → no. of ambident ligand

→ **Solvate/hydrate isomerism**: water involved as solvent

eg: $\text{CrCl}_3 \cdot 6\text{H}_2\text{O} \rightarrow [\text{Cr}(\text{H}_2\text{O})_6]^{+3}\text{Cl}_3^-$ ✓ (violet)

• $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]^{+2}\text{Cl}_2 \cdot \text{H}_2\text{O}$ ✓ (grey-green)

• $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]^+\text{Cl} \cdot 2\text{H}_2\text{O}$ ✓

• $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3]^0 \cdot 3\text{H}_2\text{O}$ ✗ (b/c neutral complex can't attract water molecules)

→ **Coordination isomerism**: interchanging of ligand b/c cationic & anionic complex.

eg: $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$, $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$

$[\]^+ [\]^-$

② Stereo isomerism

→ **Geometrical isomerism**

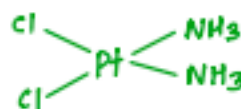
• Sq. planar ✓, tetrahedral ✗, octahedral ✓

b/c relative position of unidentate ligand w.r.t central metal ion are same

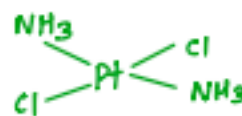
* In Heterolaptic complex

• **Cis**: two same ligand adj to each other

• **trans**: two same ligand opp to each other



Cis-platin
(anti cancerous)



trans-platin

→ **Optical isomerism**

• Sq. planar ✗, tetrahedral ✓, octahedral ✓

b/c p-o-s present

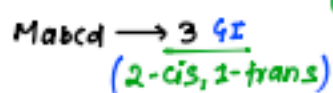
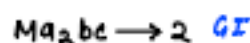
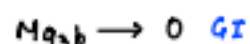
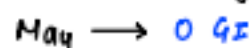
• cis ✓

• trans ✗

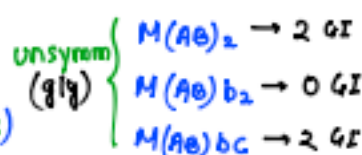
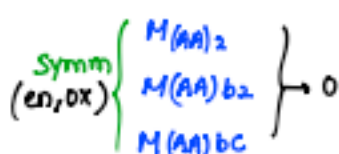
• non-superimposable mirror img (enantiomers)

(i) **Sq. planar** (CN=4) (dsp^2) (GI^V, OI^X)

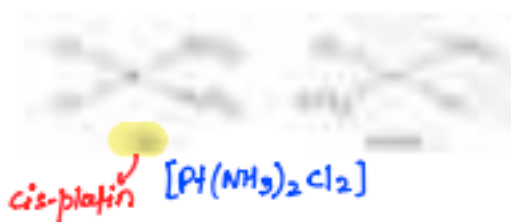
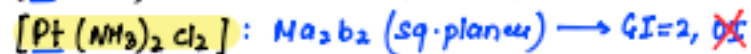
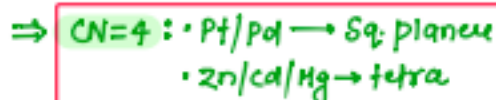
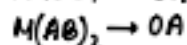
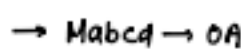
Monodentate ligand (a)



Bidentate (AA)



(ii) **Tetrahedral** (CN=4) (sp^3) (GI^X, OI^V)



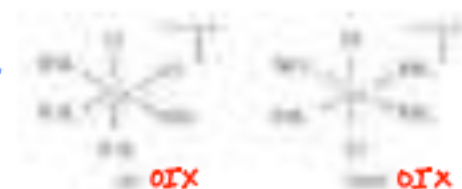
(iii) **Octahedral** (CN=6) (GI^V, OI^V)

- Trans $\rightarrow$ if two same ligand present at any one trans position: **optically inactive** (b/c pos present)
- Cis + (atleast-1-bidentate ligand) $\rightarrow 0A$ (2 same ligand adj to each other)
- 3 Rings $\rightarrow 0A$

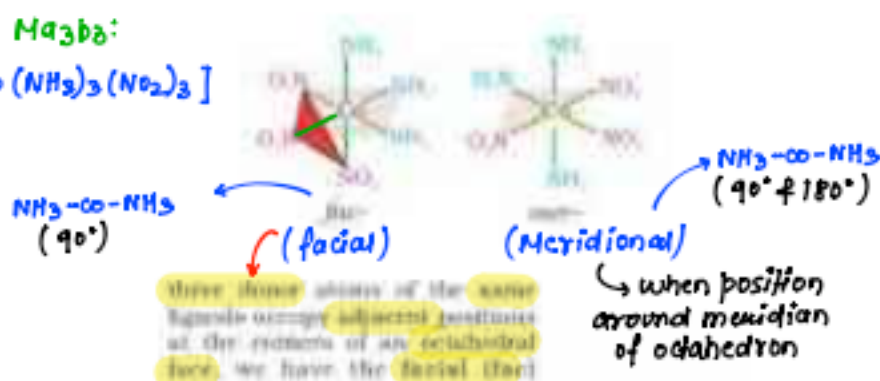
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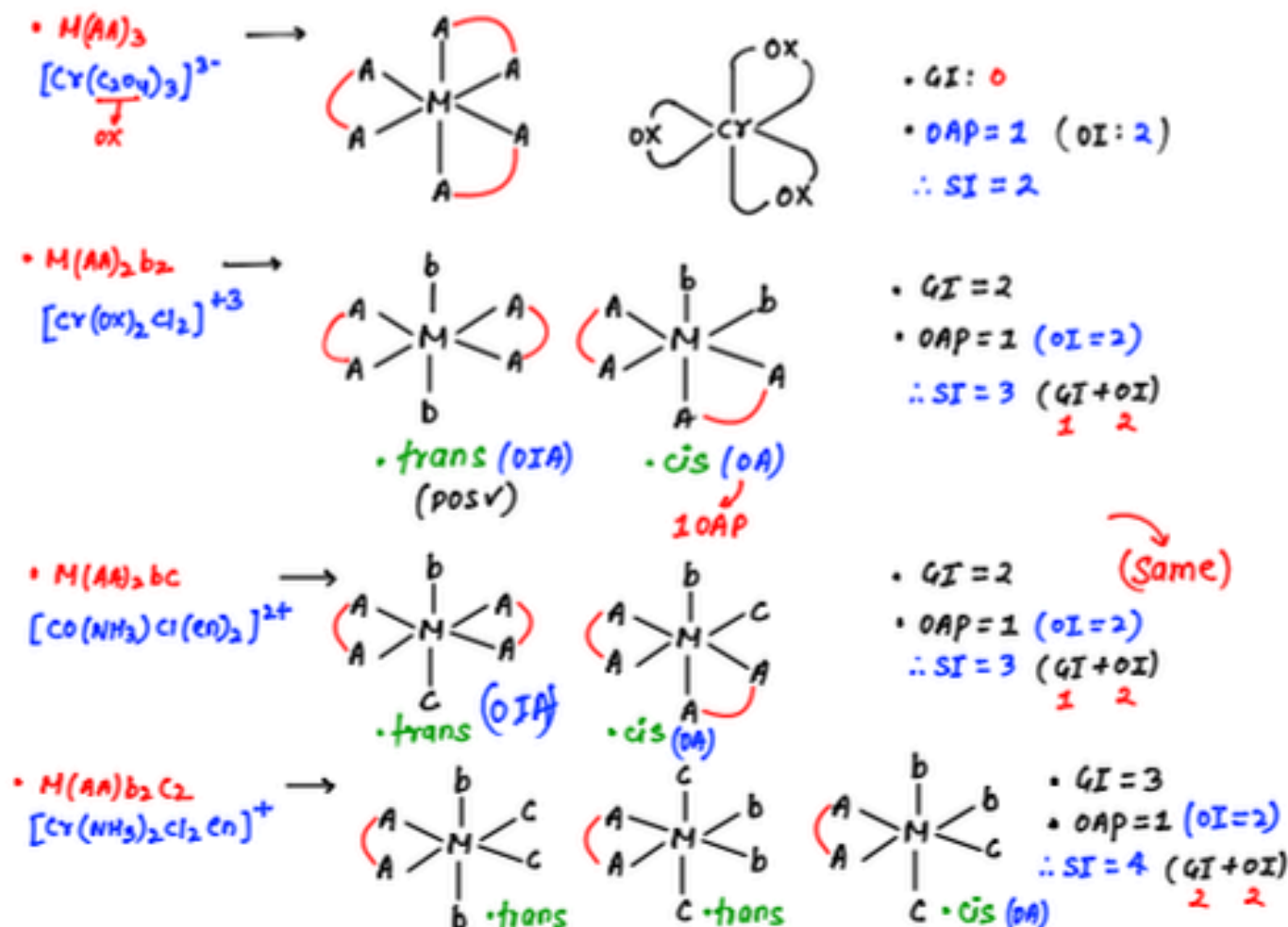
- $Ma_6 \rightarrow GI=0, OAP=0$
- $Ma_4b_2 \rightarrow GI=2, OAP=0$
- $Ma_3b_3 \rightarrow GI=2, OAP=0$
- $M(AA)_3 \rightarrow GI=0, OAP=1 (\therefore SI=2)$
- $M(AA)_2b_2 \rightarrow GI=2, OAP=1 (\therefore SI=3)$
- $M(AA)_2bc \rightarrow GI=2, OAP=1 (\therefore SI=3)$
- $M(AA)b_2c_2 \rightarrow GI=3, OAP=1 (\therefore SI=4)$

* Ma_4b_2 :



* Ma_3b_3 :





⑫ Importance of coordination compounds

- Coordination compounds are used in many applications and are very important in many fields. They are used in many ways, such as in the treatment of diseases, in the extraction of metals, and in the synthesis of new materials.
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- Chemotherapy compounds are used as **antimetabolites** for cancer treatment. Examples include **methotrexate**, **5-fluorouracil**, and **azathioprine**. In most cases, the target enzyme is a **carbamoyl transferase**.
- Methotrexate is **antimetabolite** with **glutamate** and **glutamine** metabolism and affects both synthesis of the purines **UMP** and **UMP**. There is a reduction of **UMP** levels.
- In **acute and chronic leukaemia**, the **antimetabolite** is used to **induce remission** and **prevent relapse** while disease. The **antimetabolite** **5-fluorouracil** is used to **induce remission** and **prevent relapse**.
- There is growing interest in the use of **antimetabolites** in cancer treatment. An example is the treatment of patients with the **antimetabolite** **5-fluorouracil** in the **prevention of relapse** after the **remission** of **acute and chronic leukaemia**. **5-fluorouracil** is used in the **prevention of relapse** after the **remission** of **acute and chronic leukaemia**. **5-fluorouracil** is used in the **prevention of relapse** after the **remission** of **acute and chronic leukaemia**.

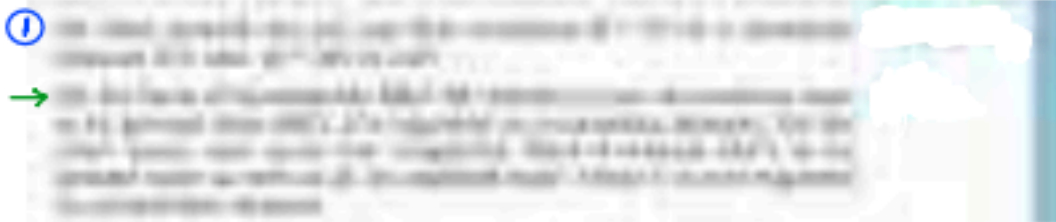
→ anti-cancerous: inhibit tumour

D & F BLOCK

* D-Block

① Transition metals: Incomplete 'd subshell' in neutral/ions state
(of existed os)

- partially filled d: $d^1 \rightarrow d^9$: Transition element
 - complete filled d: d^0, d^{10} : Non-transition
- } D-block
- ↪ Zn, Cd, Hg



* Electronic conf

- imp: 3d series
- exceptions
 - Pd, Pt, W

3d-series

Scandium	Sc	21	$[Ar] 3d^1 4s^2$
Titanium	Ti	22	$[Ar] 3d^2 4s^2$
Vanadium	V	23	$[Ar] 3d^3 4s^2$
Chromium	Cr	24	$[Ar] 3d^5 4s^1$
Manganese	Mn	25	$[Ar] 3d^5 4s^2$
Iron	Fe	26	$[Ar] 3d^6 4s^2$
Cobalt	Co	27	$[Ar] 3d^7 4s^2$
Nickel	Ni	28	$[Ar] 3d^8 4s^2$
Copper	Cu	29	$[Ar] 3d^{10} 4s^1$
Zinc	Zn	30	$[Ar] 3d^{10} 4s^2$

Sun Tu Vivah Crle Mujhse Fer Coi Nahi Cahega Zanu

5d-series

Hafnium	Hf	72	$[Xe] 4f^{14} 5d^2 6s^2$
Tantalum	Ta	73	$[Xe] 4f^{14} 5d^3 6s^2$
Tungsten	W	74	$[Xe] 4f^{14} 5d^4 6s^2$
Rhenium	Re	75	$[Xe] 4f^{14} 5d^5 6s^2$
Osmium	Os	76	$[Xe] 4f^{14} 5d^6 6s^2$
Iridium	Ir	77	$[Xe] 4f^{14} 5d^7 6s^2$
Rhodium	Rh	78	$[Xe] 4f^{14} 5d^8 6s^1$
Palladium	Pd	79	$[Xe] 4f^{14} 5d^{10}$
Silver	Ag	79	$[Xe] 4f^{14} 5d^{10} 6s^1$
Mercury	Hg	80	$[Xe] 4f^{14} 5d^{10} 6s^2$

4d-series

Yttrium	Y	39	$[Kr] 4d^1 5s^2$
Zirconium	Zr	40	$[Kr] 4d^2 5s^2$
Niobium	Nb	41	$[Kr] 4d^4 5s^1$
Molybdenum	Mo	42	$[Kr] 4d^5 5s^1$
Technetium	Tc	43	$[Kr] 4d^5 5s^2$
Ruthenium	Ru	44	$[Kr] 4d^7 5s^1$
Rhodium	Rh	45	$[Kr] 4d^8 5s^1$
Palladium	Pd	46	$[Kr] 4d^{10}$
Silver	Ag	47	$[Kr] 4d^{10} 5s^1$
Cadmium	Cd	48	$[Kr] 4d^{10} 5s^2$

La Hafta Tau Warna Re Os Iron rod se Pitai Aur Hogi

Ye Zindagi Nahi Mohabbat Tumhari Ru Rokr Pukadegi Ag Chandni

② Properties:

- High tensile strength, ductility, Malleability
- High thermal & electrical conductance, Metallic lustre
- Low volatile $\therefore$ High BP & MP
- Very Hard (except = Zn, Cd, Hg)

except: they have one/more metallic str at normal Temp^r.
Zn, Cd, Hg, Mn

③ Atomic size

* $Sc > Ti > V > Cr < \textcircled{Mn} > Fe \approx Co \approx Ni < Cu < Zn$ $\left(\begin{array}{l} \bullet Sc > Zn \\ \bullet Mn \approx Zn \end{array} \right)$

* 3rd-grp 4th → 12th grp

Sc
↑
Y
↑
La

3d
↑
4d
↑
5d

lanthanide contraction (poor shielding of 4f)

{ due to LC, Zr & Hf have similar radii & have similar physical & chemical prop. }

* Ti → Cu: d ↑
B/C: R ↓, M ↑ ($J = \frac{M}{V}$)

④ Ionisation energy

• $Sc < < < Zn$

• $Y < < < Cd$

• $La < < < Hg$

(min)

(max)

$IE \propto \frac{1}{Size}$

* 3d-series:

d^1s^2 d^3s^2 d^5s^1 d^2s^2 d^5s^2 d^8s^2 $d^{10}s^1$
Sc < Y < Cr < Ti < Mn < Ni < Cu
(Suno) (Vinita) (Cama) (Hw) (Mani) (Nahi) (Cy)

Co < Fe < Zn
(Choro) (Fir) (Zane) d⁰
 d^7s^2 d^8s^2 $d^{10}s^2$

⑤ imp points:

IE₁: Cr < Mn
 $3d^5 4s^1$ $3d^5 4s^2$

Mn < Fe
 $3d^5 4s^2$ $3d^6 4s^2$

IE₂: Cr⁺ > Mn⁺
 $3d^5$ $3d^5 4s^1$

Mn⁺ < Fe⁺
 $3d^5 4s^1$ $3d^6 4s^1$

IE₃: Cr²⁺ < Mn²⁺
 $3d^4$ $3d^5$

Mn²⁺ > Fe²⁺
 $3d^5$ $3d^6$

⑥ IE in groups

3s 3d 4d
↓ ↑ ↑
Y 4d 3d
↓ ↑ ↑
La 5d 5d
(3) (4,5,6,10) (Remaining)

* Except 3rd group, IE maximum in 5d series

⑦ Spin only magnetic moment

$\mu = \sqrt{n(n+1)}$
no of unpaired e⁻

$\left\{ \begin{array}{l} n=2 \\ \therefore \mu=2.8 \end{array} \right\}$

• $n=0$: diamagnetic

• $n \neq 0$: paramagnetic

⑧ Alloy formation (due to similar size)

- Homogeneous solid soln
- Radii about 15% of each other
- Hard & High MP

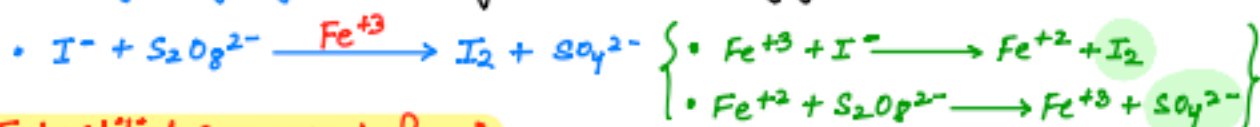
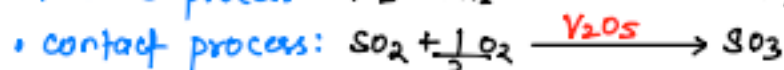
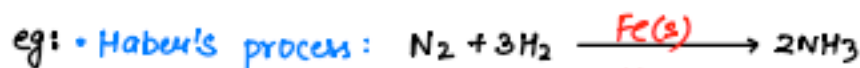
eg: • Ferrous alloys: Cr, Ni, W, Mn, V
(best known) (used in stainless steel)

• Brass: $\frac{Cu}{d} + \frac{Zn}{d}$ } transition element
• Bronze: $\frac{Cu}{d} + \frac{Sn}{p}$ } with non-transition element

③ Catalytic properties Ea ↓

- b/c of their variable oxidation state
- complex formation tendency

→ increase concⁿ of reactant at surface & weakening the reacting molecule bond



④ Interstitial compound formⁿ

• trapping small atoms → BC NH

• Non-stoichiometric compound: (Not follow law of multiple proportion)

→ TiC, Mn₄N, Fe₃H, VH_{0.56}, TiH_{1.7}

• Neither ionic, nor covalent

• Hardness ↑ (NiB: Hardness similar to diamond)

- Melting point ↑
- Conductance remain same
- Chemically inert.

⑤ Melting points

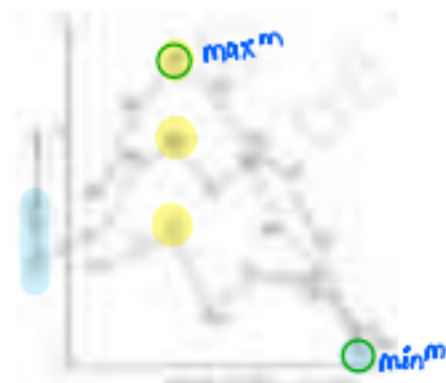
• unpaired 'd' electron

↓
Strong interatomic interaction (Bonding)

↓
High enthalpy of atomisation (ΔH_a)

∴ High MP & BP

→ indicates that metal tend to Nobel in their Reaction



* 3rd 12th
3d
4d
5d

4 → 10
3d
4d
5d
(Normal order)

11
3d
4d
5d
(3d > 5d > 4d)

* 6th group

Cr
↑
Mo
↑
W
(Max^m MP)

11th group

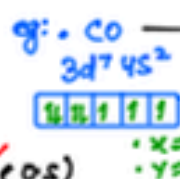
Cu
↑
Ag
↑
Au
(Min MP)

(Max^m MP) ← in period → (Min MP)

⑥ Variable oxidation state

⑧ → low energy gap b/w $(n-1)d$ & ns orbitals

* OS Range → (x) to $(x+y)$
 e^- of ns^0 unpaired e^- of $(n-1)d$



Mn show Highest OS Range
 $Mn \rightarrow 3d^5 4s^2$
 $x=2$
 $y=5$

* SC & Zn → show only 1 type OS (variable OS)
 $(+3)$ $(+2)$

* $Sc^{+2}, Fe^{+5}, Co^{+5}, Cr^{+1}$ } Not exist * stability: $Ti^{+4} > Ti^{+3} \neq Ti^{+2}$

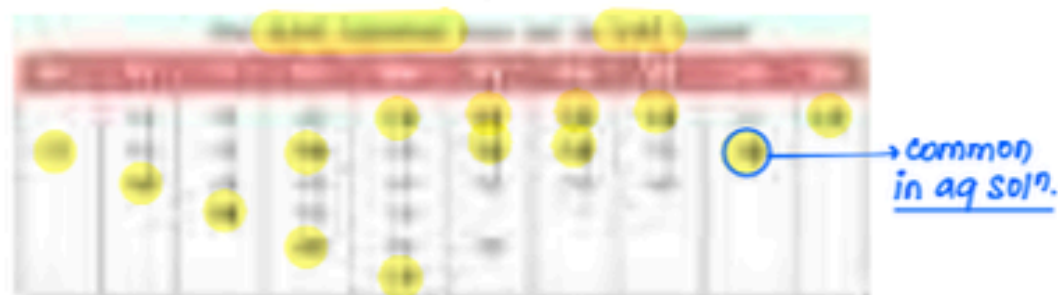
* Less number of O.S at extreme ends

SC, Ti
 (few e^- to lose)

Cu, Zn
 (too many $e^- \therefore$ less available orbital in which e^- share)



* $SC \xrightarrow[OS \text{ Range } \uparrow]{\text{upto Mn}} Mn \xrightarrow[OS \text{ Range } \downarrow]{\text{after Mn}} Zn$

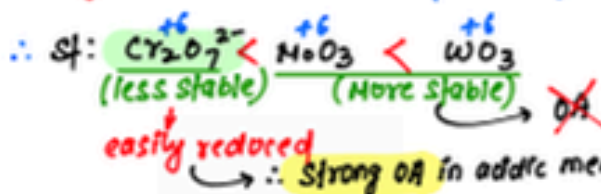
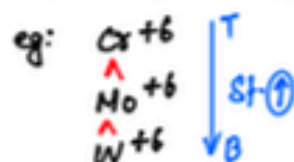


* variability of OS in
 → Transition element: differ by ① ($V^{+2}, V^{+3}, V^{+4}...$)
 due to incomplete filling of 'd-orbital'
 → Non-Transition element: differ by ②

* Along group: (In d-block)

($T \rightarrow b$)

Stability of H.O.S ① (opp of inert pair effect)



$\therefore$ Strong OA in acidic medium

* In complex compound:

- non-classical ligand/
π-acceptor/
synergic bonding



low OS are found

eg: $Ni^0(CO)_4$, $Fe^0(CO)_5$

a) Halides

* Higher OS: Ti^{+4} , V^{+5} , Cr^{+6}

• MnF_2, MnF_3, MnF_4 (exist), MnF_5, MnF_6, MnF_7 (Not exist)

↓
due to steric hindrance

{ Mn^{+7} Not exist, but
 MnO_3F exist. }

* Fluorides has ability to stabilise H.O.s due to

High lattice energy (in ionic)
• CoF_3

High bond energy (in covalent)
• VF_5, CrF_6

* V^{+5} only stable with VF_5 (other VX_5 : unstable)
($X=Cl, Br, I$)

* ∴ Fluorides stabilise HOS

∴ Fluorides are unstable in lower OS (VX_2, CX_3)

VF_2 (unstable)

Hydrolysis/decompose → oxohalides
(VOX_3)

{ eg: $VCl_5 \rightarrow VOCl_3$ }

* $CuF_2, CuCl_2, CuBr_2, CuI_2$



* Normally: $Cu^+ > Cu^{+2}$

In aq. medium: $Cu^{+2}(aq) > Cu^+(aq)$

{ $Cu^+ + Cu^+ \xrightarrow{\text{disproportionation}} Cu^{+2} + Cu$ }

Ⓡ • More -ve Hydration energy
(-OH released)
more than IE_2 of Cu Req

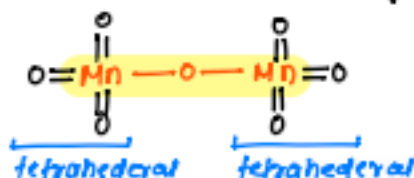
b) Oxides

• HOS of d-block element stabilise by $F \neq O$ (due to High EN)

* 'O' has more ability to stabilise HOS, due to multiple bond form tendency

* eg: • $Mn^{+4}F_4$, $Mn^{+7}_2O_7$

• covalent green oil, explosive



* Except **SC**: all metals form ' M^{+2} ' oxide (ionic)

→ B/C: $ScO \rightarrow Sc^{+2}$ (Not exist)

* HOS consider with group number but from Sc_2O_3 ($grp:3$) to Mn_2O_7 ($grp:7$)

* HOS of (Fe) is ' Fe^{+3} ': Fe_2O_3 , but FeO_4^{2-} (but only in alkaline medium for short time)

* Some stable: VO_2^+ , VO^{2+} , TiO^{2+} (after that: $FeO_4^{2-} \xrightarrow{\text{decompose}} Fe_2O_3 + O_2$)

* $OS \propto \text{covalent chr} \propto \frac{1}{\text{ionic chr}} \propto \frac{1}{MP}$ { $\therefore Mn_2O_7$: covalent green oil
 CrO_3, V_2O_5 : low MP }

$OS \propto \text{Acidic chr} \propto \frac{1}{\text{Basic chr}}$

of Basic strength

• CrO ($basic$) $>$ Cr_2O_3 ($amphoter$)
 • V_2O_3 ($basic$) $>$ V_2O_4 ($less basic$) $>$ V_2O_5 ($amphoter$)
 but mainly acidic

• $V_2O_4 \xrightarrow{\text{Acid}} VO^{+2}$

• $V_2O_5 \xrightarrow{\text{Alkali}} VO_4^{3-}$
 $\xrightarrow{\text{Acids}} \cancel{VO_4^{+2}}, VO_2^{+5}$



⑤

→

⑥

⑦ Std. Reduction potential (SRP): E° $E^\circ = -ve$: NS
 $E^\circ = +ve$: S

• $E^\circ \text{M}^{2+}/\text{M} = -ve$ (Non spontaneous)

• $E^\circ \text{Cu}^{2+}/\text{Cu} = +ve$ (Spontaneous)

* Cu can't liberate H_2 gas from acids



$\text{Cu(s)} \nrightarrow \text{Cu}^{2+}(\text{aq})$ $\rightarrow$ B/C: High energy of transposition
 not balanced by its ΔH_{hyd}

* but 'Cu' can be oxidised by only oxidising acids $\rightarrow \text{C.HNO}_3$ & Hot $\text{C.H}_2\text{SO}_4$



* $E^\circ \text{M}^{3+}/\text{M}^{2+}$

E°

Reason

• $\text{Sc}^{3+}/\text{Sc}^{2+}$: low (-ve)

$\rightarrow \text{Sc}^{3+}$: Noble gas config

• $\text{Zn}^{3+}/\text{Zn}^{2+}$: High (+ve)

$\rightarrow \text{Zn}^{2+}$: d^{10} config

• $\text{Mn}^{3+}/\text{Mn}^{2+}$: High (+ve)

$\rightarrow \text{Mn}^{2+}$: d^5 config

• $\text{Fe}^{3+}/\text{Fe}^{2+}$: comparative low (-ve)

$\rightarrow \text{Fe}^{3+}$: d^5 config

• $\text{V}^{3+}/\text{V}^{2+}$: comparative low (+ve)

$\rightarrow \text{V}^{2+}$: t_{2g}^3 level

* less -ve E° value $\xrightarrow{\text{Related}}$ inc. in $\text{IE}_1 + \text{IE}_2$
 (M^{2+}/M)

but, E° value: $\text{Mn}, \text{Zn}, \text{Ni} \rightarrow$ more -ve than expected

(M^{2+}/M)

Stability

$(\text{Mn}^{2+}: d^5)$
 $(\text{Zn}^{2+}: d^{10})$

Highest -ve ΔH_{hyd}

* SRP depend on $\rightarrow$

- Atomisation enthalpy (ΔH_a)
- Ionisation energy (IE)
- Hydration enthalpy (ΔH_{hyd})

* $E^\circ \text{M}^{3+}/\text{M}^{2+}$:

• $\text{Ti}^{3+}/\text{Ti}^{2+} = -ve$

• $\text{V}^{3+}/\text{V}^{2+} = -ve$

• $\text{Cr}^{3+}/\text{Cr}^{2+} = -ve$

• $\text{Co}^{3+}/\text{Co}^{2+} = +ve$

• $\text{Mn}^{3+}/\text{Mn}^{2+} = +ve$

$\therefore \text{Ti}^{2+}, \text{V}^{2+}, \text{Cr}^{2+}$: Strong RA & liberate Hydrogen from dil. Acid.
 $(\text{Cr}^{2+} + \text{H}^+ \rightarrow \text{Cr}^{3+} + \text{H}_2)$

$\therefore \text{Mn}^{3+} \& \text{Co}^{3+}$: strong OA in aq. sol

⑦

$\rightarrow$

⑧

⑨

→

⑩

→

$d^4 \rightarrow d^5$

⑪

⑫

⑬ Calculate the magnetic moment of a divalent ion in aqueous solution if its atomic number is 25.

→ With atomic number 25, the divalent ion in aqueous solution will have d^3 configuration (five unpaired electrons). The magnetic moment, μ is

$$\mu = \sqrt{5(5+2)} = 5.92 \text{ BM}$$

Interest Question

⑭ Calculate the spin only magnetic moment of M^{2+} ion (Z = 27).

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Atomic number	21	22	23	24	25	26	27	28	29	30
Electronic configuration										
3d	$3d^1 4s^2$	$3d^2 4s^2$	$3d^3 4s^2$	$3d^5 4s^1$	$3d^5 4s^2$	$3d^6 4s^2$	$3d^7 4s^2$	$3d^8 4s^2$	$3d^{10} 4s^1$	$3d^{10} 4s^2$
3d	$3d^2 4s^2$	$3d^2 4s^2$	$3d^3 4s^2$	$3d^4 4s^1$	$3d^5 4s^2$	$3d^6 4s^2$	$3d^7 4s^2$	$3d^8 4s^2$	$3d^9 4s^1$	$3d^{10} 4s^2$
3d	$3d^1 4s^2$	$3d^2 4s^2$	$3d^3 4s^2$	$3d^4 4s^1$	$3d^5 4s^2$	$3d^6 4s^2$	$3d^7 4s^2$	$3d^8 4s^2$	$3d^9 4s^1$	$3d^{10} 4s^2$
3d	$3d^1 4s^2$	$3d^2 4s^2$	$3d^3 4s^2$	$3d^4 4s^1$	$3d^5 4s^2$	$3d^6 4s^2$	$3d^7 4s^2$	$3d^8 4s^2$	$3d^9 4s^1$	$3d^{10} 4s^2$
Enthalpy of atomization, $\Delta H^\circ_{\text{at}}$ (kJ mol ⁻¹)	336	473	515	607	681	740	815	891	1031	1136
Ionization enthalpy, $\Delta H^\circ_{\text{ion}}$ (kJ mol ⁻¹)										
1st	611	658	659	603	717	742	759	736	743	900
2nd	1251	1309	1414	1558	1509	1561	1549	1752	1958	1733
3rd	2202	2352	2552	2966	3251	3390	3553	4470	5059	3827
Melting/boiling point	1538	1668	1910	1907	1744	1538	1495	1455	1357	1357
Standard electrode potential E° (V)										
M^{2+}/M	-	-1.63	-1.18	-0.90	-1.18	-0.44	-0.28	-0.25	-0.52	-0.76
M^{3+}/M^{2+}		-0.37	-0.26	-0.41	-0.45	-0.77	-1.82	-	-	-
Density/g cm ⁻³	2.98	4.54	6.01	7.19	7.43	7.86	8.90	8.90	8.96	7.14

→ $\Delta H_{\text{at}} \propto \text{unpaired } e^-$

ΔH_{at} : $\bullet V$: Highest
 $\bullet Zn$: lowest
 (3d series)

⑮

→

⑯

⑧ colour of ions

Ⓐ: 'd-d' transition & charge transfer

* d^0, d^{10} : colourless
 $(Sc^{+3}, Zn^{+2}, Cu^{+1}, Ag^{+1})$

* d^{1-9} : coloured ✓

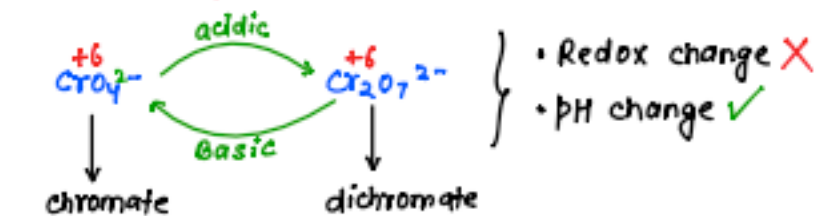
⑨ complex formⁿ tendency

- Small size
- High ϕ nature (covalent chr ↑)
- Vacant orbital

Electron configuration	Complex formation	Colour
d^0	Weak	Colourless
d^{1-9}	Strong	Coloured
d^{10}	Weak	Colourless



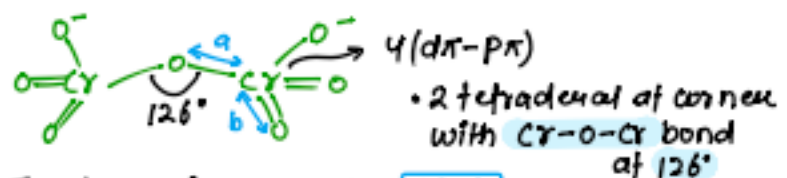
⑩ $K_2Cr_2O_7$: potassium dichromate



* coloured due to charge transfer (LMCT)

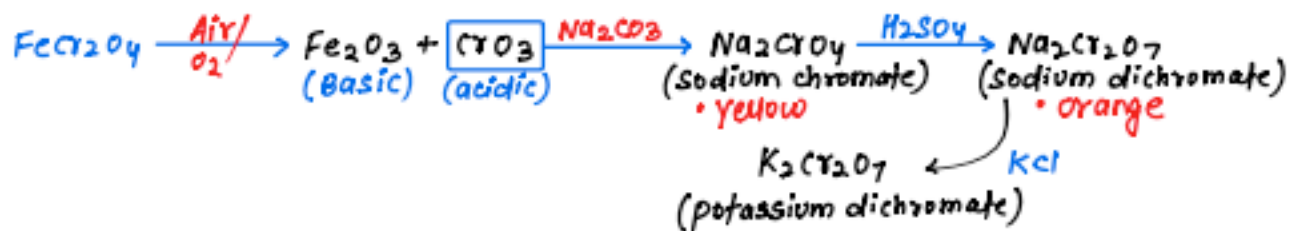
- yellow
- Stable in basic medium
- CrO_4^{2-} is tetrahedral
 $2(d\pi-p\pi)$
- Avg BO: $(Cr-O) \rightarrow 6/4$
- $\therefore$ it is resonance hybrid
 $\therefore$ all bond length are equal

- orange
- Stable in acidic medium



- Two types of bond length: $a > b$
- 6 $(Cr-O)$ terminal bonds $\rightarrow$ equal due to resonance

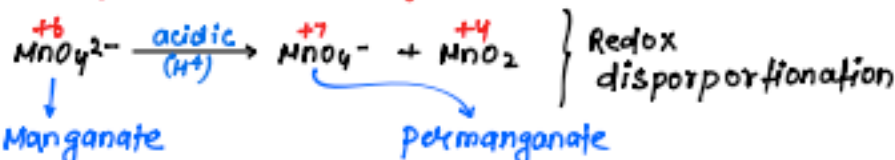
* Preparation (by chromate ore: FeCr_2O_4)



* $\text{Na}_2\text{Cr}_2\text{O}_7$ & $\text{K}_2\text{Cr}_2\text{O}_7 \rightarrow$ both: strong OA.

- more soluble in H_2O (Hygroscopic)
- extensively used as OA in organic chemistry
- used as primary standard in volumetric analysis

① KMnO_4 : potassium permanganate (KMnO_4)



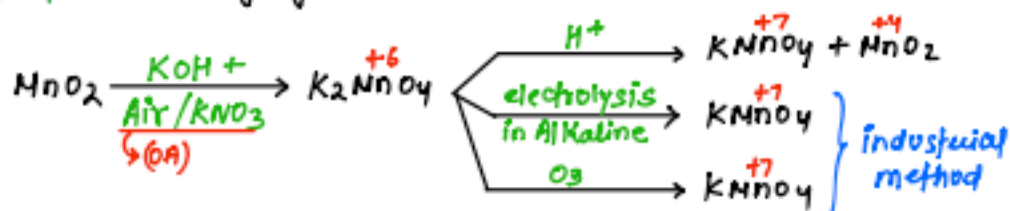
* coloured due to charge transfer (LMCT)

- | | |
|---|---|
| <ul style="list-style-type: none"> • dark green • stable in basic medium • $\text{O}^- - \text{Mn} = \text{O} \rightarrow 2(d\pi - p\pi)$
• Tetrahedral • Avg BO: (M-O): $6/4 (=1.5)$ • Due to Resonance all bond length are equal • <u>less BO & BE, more BL</u> | <ul style="list-style-type: none"> • purple • stable in Acidic medium • $\text{O}^- - \text{Mn} = \text{O} \rightarrow 3(d\pi - p\pi)$
• tetrahedral • Avg BO: (M-O): $7/4 (=1.75)$ • Due to Resonance all bond length are equal • <u>more BO & BE, less BL</u> |
|---|---|

* Mn^{+6} : d^1 ($n=1$)
↓
paramagnetic

* Mn^{+7} : d^0 ($n=0$)
∴ diamagnetic (normally)
{ but they are temp^s dependent }
weak paramagnetic.
→ MDT

* Preparation (by pyrolusite ore: MnO_2)

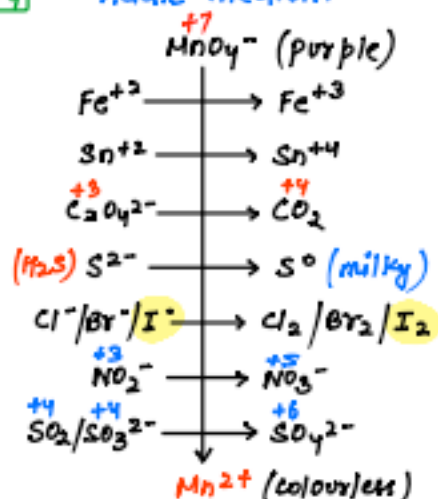


(12) Important properties

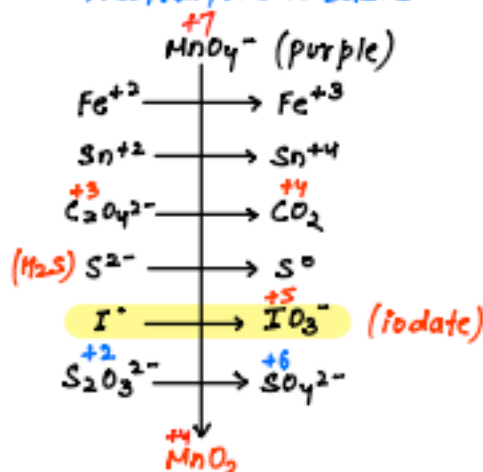
* OA: $\text{MnO}_4^- > \text{Cl}_2 > \text{Cr}_2\text{O}_7^{2-} > \text{VO}_2^+$

KMnO₄

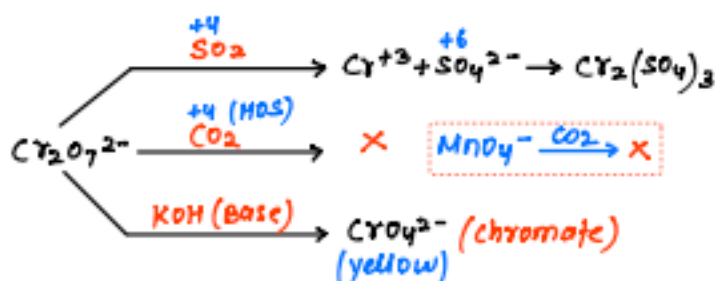
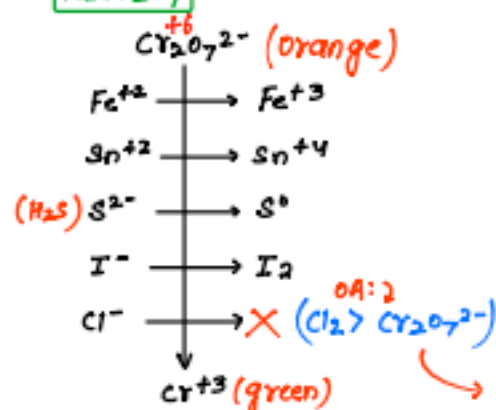
Acidic medium



Neutral/weak basic



K₂Cr₂O₇



HCl can't be used as acidic medium in KMnO₄ titration but can be used in K₂Cr₂O₇ titration

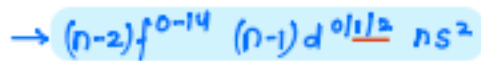
$$\text{Cl}^- \xrightarrow{\checkmark} \text{Cl}_2 \quad \text{Cl}^- \xrightarrow{\text{X}} \text{Cl}_2$$

(13) Chromyl chloride test

- Given by: Chloride ion ($\text{NaCl}, \text{KCl} \dots$)
- Not Given by: $\text{Ag}^+, \text{Cu}_2^{2+}, \text{Hg}_2^{2+}, \text{Sn}^{+4}, \text{Pb}^{+4}$



⑭ f-block



Lanthanide series 58 → 71

Cene Par Nadiya Prem se
Samai, Eu Gadgad Tb Dil
Hoa Ere Tm Ybi Lutlo

Actinide series 90 → 103

Taro Par Unki Nazar Pari
Asman Chamka Bizli CarKi
Esan Faramos Manchali Nikli
Lrki

⑮ Lanthanide

- poor σ -shielding of $4f e^-$, $Gd: (Xe) 6s^2, 4f^7, 5d^1$
- $\therefore Z_{eff} \uparrow$: contraction $\uparrow$, size $\downarrow$
(Lanthanide contraction)

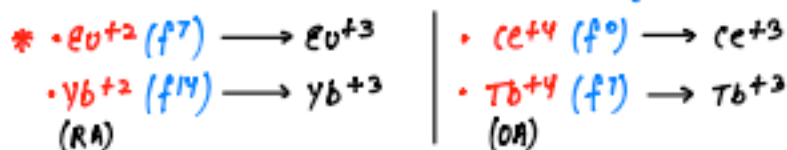
- Similar size of Zr & Hf $\rightarrow$ they have similar physical & chemical prop.
- Common o.s of Lanthanide: Ln^{+3}
- +2 & +4 ions obtained b/c of Half/full/noble confⁿ



- but: $Ce^{+4} \xrightarrow{(OA)} Ce^{+3} \therefore E^\circ_{Ce^{+4}/Ce^{+3}} : +ve$

$\rightarrow$ thermodynamically it oxidise water

$\rightarrow$ but, kinetically reaction rate very slow $\therefore Ce^{+4}$ used as analytical reagent



⑨ Lanthanide characteristics

• Soft metal, silvery, conductors

• $L \rightarrow R$: Hardness $\uparrow$ (Sm: Steel hard)
 $\rightarrow$ Highest MP

• Density variation: Eu, Yb, Sm, Tm } in others, density & other prop. smoothly
 Sometimes

* All trivalent (Ln^{+3}) $\rightarrow$ coloured (due to f^1e^-) } except: La^{+3}, Lu^{+3} (colourless)

* $La^{+3} \rightarrow f^0$ $Yb^{+2} \rightarrow f^{14}$ $\therefore n=0$
 $Ce^{+4} \rightarrow f^0$ $Lu^{+3} \rightarrow f^{14}$ $\therefore$ Diamagnetic

* Ionisation Energy of Ln:

• $IE_1 = 600 \text{ kJ/mol}$
 • $IE_2 = 1200 \text{ kJ/mol}$ } these value similar to calcium

• earlier members are quite reactive & similar to calcium.

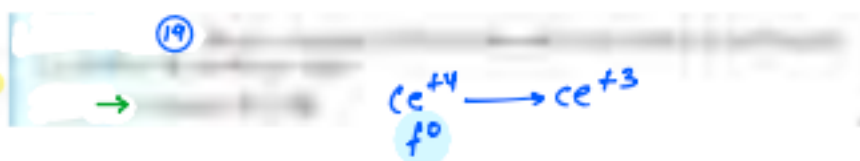
• with $AN \uparrow$: they behave like aluminium

• $IE_3 = La, Lu, Gd \rightarrow$ abnormally low
 $\downarrow \quad \downarrow \quad \downarrow$
 $La^{+3} \quad Lu^{+3} \quad Gd^{+3}$
 $(f^0) \quad (f^{14}) \quad (f^7)$

* Variation of 3rd IE indicates Exchange enthalpy

⑩ Chemical Rxn:

* **Mischmetall**: Ln (95%), Fe (5%), "S, C, Ca, Al"
 (metal alloy) $\rightarrow$ used in Mg alloy in bullet, shell, lighters



⑪ Actinide

• poor σ of $5f^1e^-$ $\therefore$ $Z_{eff} \uparrow$, Actinide contraction $\uparrow$

$\rightarrow$ **AC > LC** • B/c more poor shielding of $5f^1e^-$
 $(d \sigma > f \sigma)$
 (poor)

Th: (Rn) $7s^2, 5f^0, 6d^2$

Cm: (Rn) $7s^2, 5f^7, 6d^1$

• Radioactive metal, silvery

• Variable O.S. $\rightarrow$
 (B/c less energy gap b/w $7s, 5f, 6d$)

O.S. $\uparrow$ from: $Th^{+4}, Pa^{+5}, U^{+6}, Np^{+7}$
 then $\downarrow$ in succeeding element.

$\Rightarrow$ Max O.S.: Np^{+7}, Pu^{+7}



* Atomic size: $\text{La} \longrightarrow \text{Lu}$
 $\text{La}^{+3} \longrightarrow \text{Lu}^{+3}$) size ↓

* Basic nature of $\text{Ln}(\text{OH})_3$

Size $\propto \frac{1}{EN} \propto \frac{1}{AS} \propto BS \therefore \text{La}(\text{OH})_3 \longrightarrow \text{Lu}(\text{OH})_3$
 $\therefore \text{Size} \downarrow \therefore BS \uparrow, AS \uparrow$

* Actinide size ↑ > Lanthanide size

$\therefore A \downarrow, BS \uparrow, EN \downarrow, IE \downarrow$

BS: $\text{Ac}(\text{OH})_3 > \text{Ln}(\text{OH})_3$ Reactive: $\text{An} > \text{Ln}$

⇒ VIPs (about Actinide)

- Str variability due to irregular metallic radii, which are greater than Ln.
- Actinide are highly reactive metals (especially when finely divided)
- Action of BP on Ac $\begin{cases} \text{gives mixture of oxide \& hydride} \\ \text{comb}^n \text{ with most 'non-metals' (at moderate Temp}^r \end{cases}$

• Actinide $\begin{cases} \xrightarrow{\text{HCl}} \checkmark \text{ attack all metals} \\ \xrightarrow{\text{HNO}_3} \checkmark \text{ slightly affected (Form}^n \text{ of protective oxide layer)} \\ \xrightarrow{\text{Alkalies}} \times \end{cases}$

* Magnetic prop. of actinide more complex than Ln.
 (Magnetic susceptibility of An roughly parallel to Ln)

PYQ

* IE of early Actinide lower than early Ln:

: B/C $\begin{cases} \rightarrow sf \text{ orbital penetrate less into inner core electron} \\ \rightarrow sf\text{-electron more effectively shielded from nuclear charge than } 4f\text{-e} \\ \quad \downarrow Z_{eff} = Z - \sigma \uparrow \\ \quad \therefore IE \downarrow \\ \rightarrow \text{outer e}^- \text{ less firmly held} \therefore \text{available for bonding in Actinide} \\ \quad \text{(loose)} \end{cases}$

⑪ Some application of d & f-block

- Coinage metals: Group 11: Ag, Au
- UK copper coin: copper-coated steel
- UK silver coin: Cu/Ni alloy
- $\text{SO}_2 \xrightarrow{\text{V}_2\text{O}_5} \text{H}_2\text{SO}_4$ (contact process)
- $n(\text{CH}_2=\text{CH}_2) \xrightarrow[\text{(polymerisation)}]{\text{Ziegler Natta catalyst}} \text{-(CH}_2\text{-CH}_2\text{)}_n$ polythene

• $\text{N}_2 + 3\text{H}_2 \xrightarrow{\text{Fe}} 2\text{NH}_3$ (Haber process)

• $\begin{matrix} R & & R \\ & \backslash & / \\ & C = C \\ & / & \backslash \\ H & & H \end{matrix} \xrightarrow{\text{H}_2, \text{Ni}} \text{R-CH}_2\text{-CH}_2\text{-R}$ (Hydrogⁿ of fats)
 unsaturated saturated

• $\text{HC}\equiv\text{CH} \xrightarrow[\text{ethyne oxidation}]{\text{PdCl}_2} \text{CH}_3\text{-}\overset{\text{O}}{\underset{\text{||}}{\text{C}}}\text{-H}$ (Wacker process)

• Ni-complex useful for polymerisation of alkynes & other organic comp^d (⊙)

• AgBr: photographic industry (⊙+w)
 $\rightarrow$ light sensitive

P BLOCK

Group 15

- NaNO_3 : Chile salt
- KNO_3 : Indian salt
- $\text{Ca}_9(\text{PO}_4)_6 \cdot \text{CaF}_2$: Fluorapatite

② Physical prop:

N	} Non metal	T	• IE ↓ • atomic size ↑ • Metallic chr ↑	* NP: N < P < As > Sb > Bi * BP: N < P < As < Sb (T → B ↑)		
P						
As					} Metalloid	B
Sb						
Bi	} Metal					

• p/AS: (dπ-dπ) ✓
 ↳ ∴ p(c2)

① Anomalous prop of Nitrogen

- $\text{N} \rightarrow \text{p}\pi\text{-p}\pi$
- $\text{N-N} < \text{P-P}$: Single N-N bond weaker than single P-P bond
 High interatomic repulsion $\rightarrow$ catenation prop. weaker in Nitrogen
- $\text{N: (p}\pi\text{-d}\pi) \times$
- $\text{P/As: (d}\pi\text{-d}\pi) \checkmark$ ligands
 $\rightarrow \therefore \text{P}(\text{C}_6\text{H}_5)_3 / \text{As}(\text{C}_6\text{H}_5)_3$

The elements of this group are **poisonous**. Nitrogen is poisonous gas while all others are solid. Metallic character increases down the group. Nitrogen and phosphorus are non-metals, whereas antimony, bismuth and arsenic are metalloids. This is due to the fact that nitrogen and phosphorus have high electronegativity. The **oxidizing power** is greater, it increases from top to bottom in the group but the reducing power increases from bottom to top. Nitrogen is highly electronegative. Except nitrogen, all the elements show allotropy.

③ Reactivity

* For Hydride: $\text{T} \rightarrow \text{B}$: Stability ↓, $\propto \text{BDE} \downarrow \therefore \text{AS} \uparrow$

$$S \propto \text{O.A} \propto \frac{1}{\text{R.A}} \propto \frac{1}{0.5} \propto \frac{1}{\text{Acidity}} \propto \text{Basicity}$$

Hydride

- $\text{E} \rightarrow \text{EH}_3$
- $\text{T} \rightarrow \text{B}$:
 $\rightarrow \text{S} \downarrow, \text{BS} \downarrow$
 $\text{AS} \uparrow, \text{RA} \uparrow$
 - NH_3 : mild RA
 - BiH_3 : strong RA
 - * MP/BP: $\text{NH}_3 > \text{PH}_3$ (H-bond)

Oxide

- $\text{E} \rightarrow \text{E}_2\text{O}_3 / \text{E}_2\text{O}_5$
- $(\text{O.S} \propto \text{A})$
 - $\text{T} \rightarrow \text{B}$: $\text{A} \downarrow \text{B} \uparrow$
 - * E_2O_3 :
 $\text{N}_2\text{O}_5 / \text{P}_2\text{O}_5$: Acidic
 $\text{As}_2\text{O}_3 / \text{Sb}_2\text{O}_3$: Ampho
 Bi_2O_3 : Basic

Halides

- $\text{E} \rightarrow \text{EX}_3 / \text{EX}_5$
- covalent: $\text{EX}_5 > \text{EX}_3$
 - $\frac{\phi\pi + \sigma\pi - \sigma\pi}{\gamma^+} \propto \frac{1}{\gamma^+}$
 ↓
 polarising power of γ^+ is greater

Metals

- with Metals they exist in ③
- $\text{Ca}_3\text{N}_2^{-3}$
- $\text{Ca}_3\text{P}_2^{-3}$
- $\text{Na}_3\text{As}^{-3}$

Group 16 (Chalcogens)

(a) Oxygen compound:

- Gypsum: $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
- Epsom salt: $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
- Baryte: BaSO_4

* Sulphur compound

- Galena: PbS
- Zinc blende: ZnS
- Copper pyrites: CuFeS_2

(b) EGE (-OH):



↘ B/C of compact nature of oxygen

* (IE) $\text{grp 16} < (\text{IE}) \text{grp 15}$ ↗ B/C stable half filled (p^3) p-orbital of grp-15

(1) Physical prop

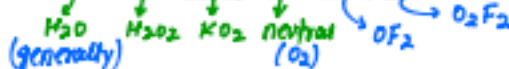
- O/S → Non-metals
- Se/Te → Metalloid
- Po → Metal
- O → O_2
- S → S_8

(2) Chemical prop:

T → B: Stability of L-O-S ↑, due to inert-pair-effect

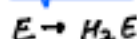
↘ ∴ +6 OS ↓, +4 OS ↑

* Oxygen show variable OS → $\text{O}^{-2}, \text{O}^{-1}, \text{O}^{-1/2}, \text{O}^0, \text{O}^{+2}, \text{O}^{+1}$

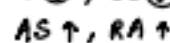


↗ H-bond present
* H_2O : liquid
 H_2S : gas

Hydride



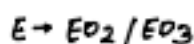
T → B (BDE ↓)



H_2O : RAX
↘ not show Reducing property

* AS: $\text{H}_2\text{S} < \text{H}_2\text{Te}$
(T → B): BDE ↓ ∴ AS ↑

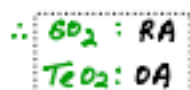
Oxide



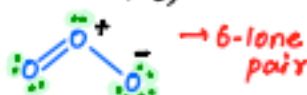
↘ acidic

- O_3 / SO_2 : gas
- SeO_2 : solid

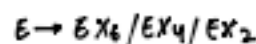
T → B: A ↓, RA ↓



* Ozone (O_3)



Halides



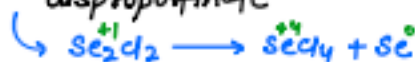
• Halides stability:



* EF_6 → only stable halide
↘ gases, octahedral

↘ SF_6 : stable for steric reason

* monohalide dismutate & disproportionate



Group 17 (Halogens)

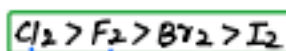
(a) Fluorspar: CaF_2

• cryolite: Na_3AlF_6

• Fluorapatite: $3\text{Ca}_3(\text{PO}_4)_2 \cdot \text{CaF}_2$

• carnallite: $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
(Double salt)

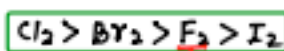
(b) EGE (-AH):



↘ Highest |EGE|

↘ B/C of smaller size of fluorine

(c) BDE:



↘ due to large e-e repulsion (F-F)

* BL: $\text{F}_2 < \text{Cl}_2 < \text{Br}_2 < \text{I}_2$

① Physical prop:

→ All Halogen coloured
↓
they absorb radiation in visible region

React with water

- F_2 (gas): yellow
- Cl_2 (gas): green-yellow
- Br_2 (liq): Red
- I_2 (solid): violet

→ sparingly react with water, but react with organic solvent

• F_2 is stronger o.a than Cl_2

→ B/c

- BDE of F_2 ↓ (due to e^-e^- repulsion)
- Hydration enthalpy of F^- ↑

② Chemical prop.

• F: exhibit only (-1) o.s
→ strongest o.a

• $T \rightarrow B$: o.a ↓ in aq. medium

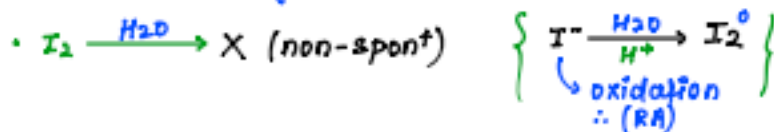
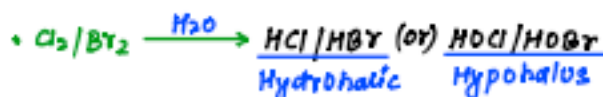
→ B/c of std. electrode potⁿ

* Halogens are highly reactive

→ $T \rightarrow B$: Reactivity ↓ & o.a ↓

• ΔH of dissociation (BDE)
• ΔH of electron gain (EGE)
• ΔH of Hydration (HE)

* Halogens reaction with H_2O

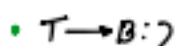


* Rxn with F_2 are exothermic

* HF : BP → (293K), liq at room temp
→ H-bond ✓

③ Reactivity

Hydride



• stability ↓

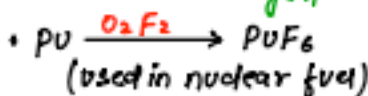
• BDE ↓

• Acidity ↑

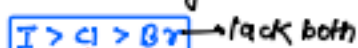
Oxide



→ Thermal stable
Strong fluorinating agent



• oxide stability



multiple bond polarisation for^{no} due to vacant d-orbital
→ By kinetic & thermodynamic factor

* Some oxide:

Cl-oxide: $Cl_2O, ClO_2, Cl_2O_6, Cl_2O_7$ } Highly reactive o.a → Tends to explode

Br-oxide: Br_2O, BrO_2, BrO_3 } least stable → powerful o.a

I-oxide: I_2O_4, I_2O_7, I_2O_5 } insoluble solids
↓
good o.a

Metal

• T → B: ionic chr ↓



• HOs Halides are more covalent



* Reactivity:



→ Reactivity of
Interhalogen > Halogen

(xx') (x₂)
High BP due to polar bond (I^{δ-}-Cl^{δ+})

④ Melting & boiling point: 15, 16, 17 Hydride

MP (112)

1 NH ₃	1 H ₂ O	2 HF
4 PH ₃	4 H ₂ S	4 HCl
3 AsH ₃	3 H ₂ Se	3 HBr
2 SbH ₃	2 H ₂ Te	1 HI

BP (211)

2 NH ₃	1 H ₂ O	1 HF
4 PH ₃	4 H ₂ S	4 HCl
3 AsH ₃	3 H ₂ Se	3 HBr
1 SbH ₃	2 H ₂ Te	2 HI

* MP & BP of NH₃, H₂O, HF: High due to H-bond

Group 18

→ called → Noble gas: B/c fully filled valance orbital
Inert gas: B/c chemically unreactive

① IE: Highest (B/c fully filled stable)

② EGE (ΔH): all have +ve ΔH value



He: lowest BP (4.2 K)

① Physical properties

* Noble gas: • Monatomic, colourless, odourless, tasteless

• Very low MP & BP: B/c they only have weak dispersion force
↓
∴ they liquefied at low Temp^r (No other Intratomic force)

② Chemical prop

A: Noble gas are least Reactive.

R: They have completely filled orbital & high IE ↑ & +ve ΔH

③ Xenon compounds (exist)

- XeF₂, XeF₄, XeF₆
- XeO₃, XeOF₄, XeO₂F₂

→ AB₂Li₁
→ XeO₃ is colourless explosive solid and has a pyramidal molecular structure
→ XeOF₄ is colourless volatile liquid and has a square pyramidal molecular structure
→ AB₅Li₁

④ Uses:

Helium

- light gas
- cryogenic agent
- NMR Spectrometer
- MRI

* Used as diluent for oxygen in diving apparatus

↓
B/c 'He' has very low solubility in blood/O₂ pyo

Neon

- Discharge tube
- Fluorescent bulb
- Green houses

Argon

- Metallurgical process
↓
for alloys / Metals making
↓
∴ It is inert at high Temp^r

* Some Imp Named Reaction

1. Haber Process (Ammonia Synthesis)



2. Ostwald Process (Nitric Acid Production)



3. Contact Process (Sulfuric Acid Production)



4. Deacon Process (Chlorine Production from HCl)

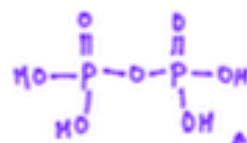


* OXYACID

per-ic acid
ic acid
ous acid
Hypo-ous acid

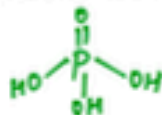
↓ OS ↓

- Meta acid: 1 Acid - 1 H₂O = MA
- pyro acid: 2 Acid - 1 H₂O = PA
- ortho acid: Same acid



① Phosphorous

phosphoric acid: H₃PO₄

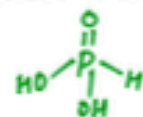


Meta phosphoric acid: H₃PO₄ - H₂O = HPO₃

pyrophosphoric acid: 2(H₃PO₄) - H₂O = H₆P₂O₇ - H₂O = H₄P₂O₇

ortho phosphoric acid: H₃PO₄

phosphorous acid: H₃PO₃



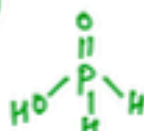
Meta phosphorous acid: HPO₂

pyrophosphorous acid: H₄P₂O₅

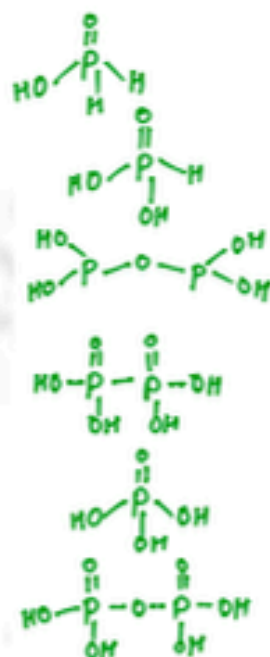
ortho phosphorous acid: H₃PO₃
(phosphonic acid)

H₄P₂O₆
(Hypophosphoric acid)

Hypophosphorous acid: H₃PO₂
(phosphinic acid)



Hypophosphorous (Phosphinic)	H ₃ PO ₂	+1	One P=O Two P-H One P-OH
Orthophosphorous (Phosphonic)	H ₃ PO ₃	+3	Two P=O One P-H One P-OH
Pyrophosphorous	H ₄ P ₂ O ₅	+3	Four P=O One P-H One P-OH
Hypophosphoric	H ₄ P ₂ O ₆	+4	Four P=O Two P-H One P-OH
Orthophosphoric	H ₃ PO ₄	+5	Three P=O One P-H One P-OH
Pyrophosphoric	H ₄ P ₂ O ₇	+5	Four P=O Two P-H One P-OH

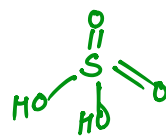


⑥ Sulphur

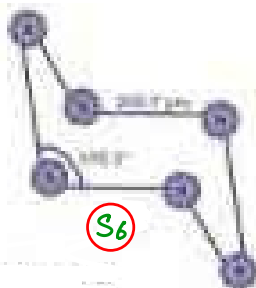
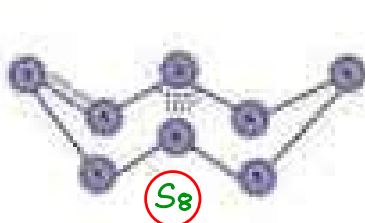
Sulphuric acid: H_2SO_4

Sulphurous acid: H_2SO_3

Pyro Sulphuric acid: $H_4S_2O_8 - H_2O = H_2S_2O_7$ (oleum)
 ortho Sulphuric acid: H_2SO_4



$H_2S_2O_8$
 (peroxodisulphuric acid)



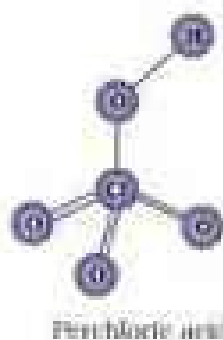
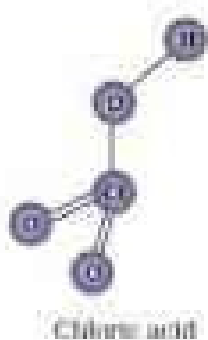
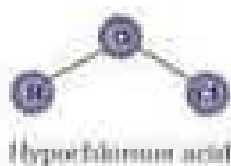
⑦ chlorine

Hypochlorous Acid: $HOCl$ ($HClO$)

Chloric Acid: $HOClO_2$ ($HClO_3$)

Chlorous Acid: $HOClO$ ($HClO_2$)

Perchloric Acid: $HOClO_3$ ($HClO_4$)



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